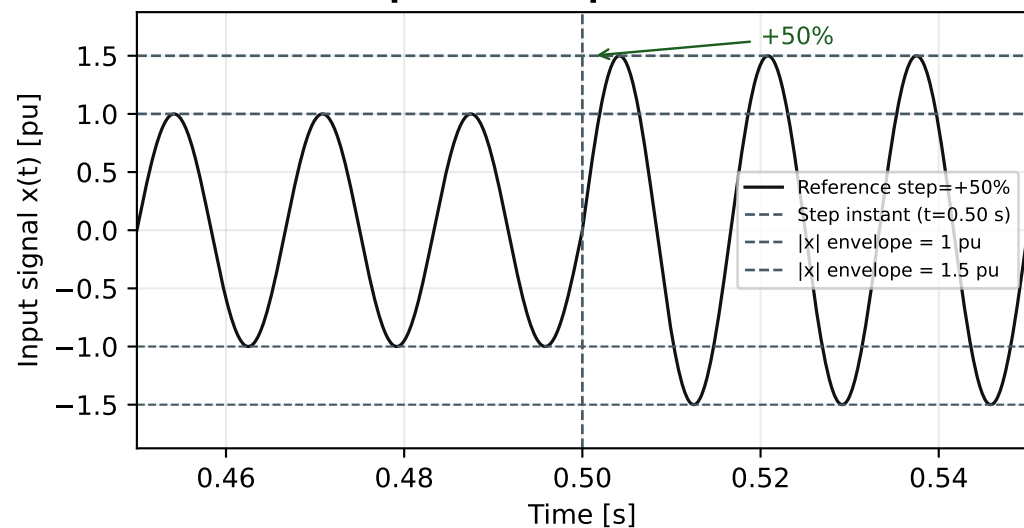


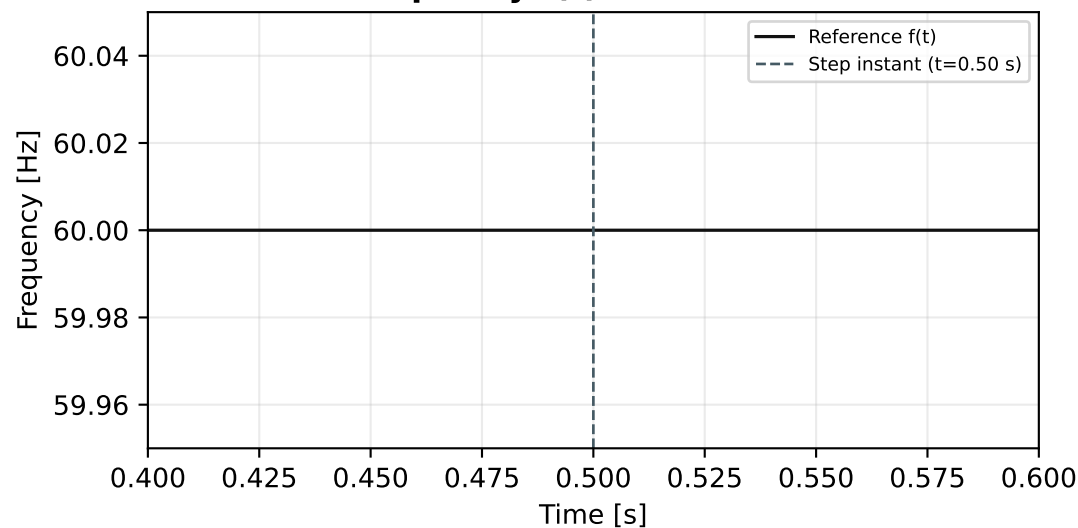
RMSE: by estimator family (Fixed Policy)

This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify magnitude-step cross-sensitivity, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits. Fixed-policy mode: one parameter set per estimator is reused across all amplitudes.

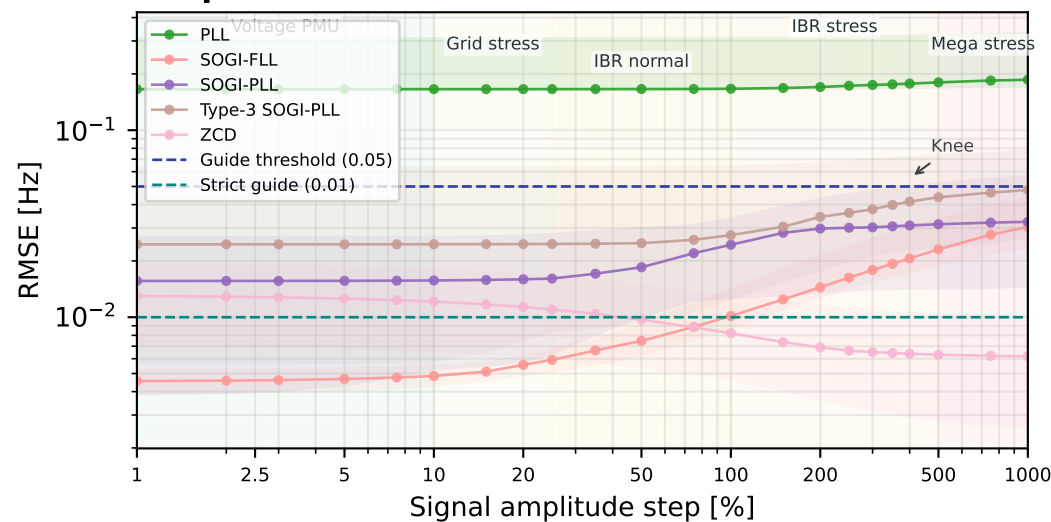
Reference Amplitude Step



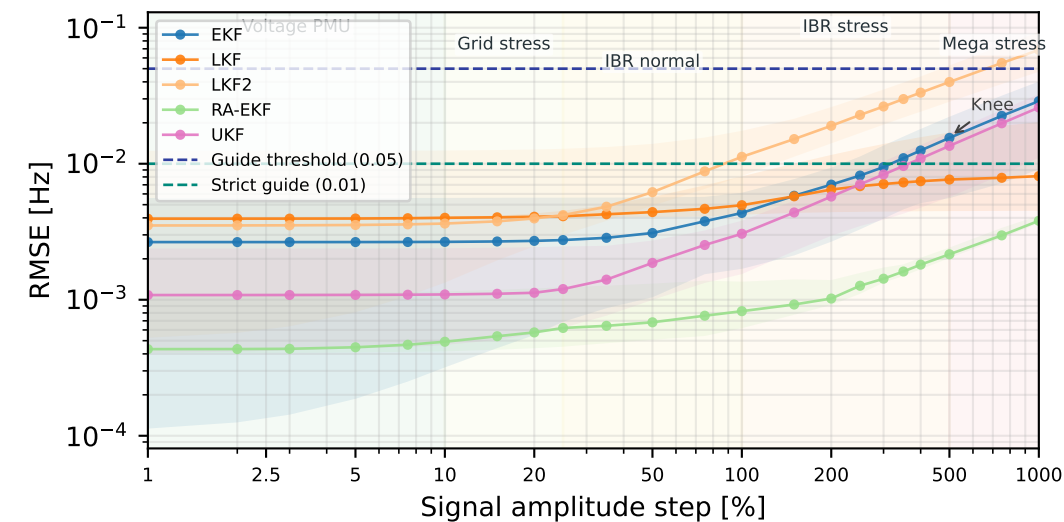
Reference Frequency $f(t)$



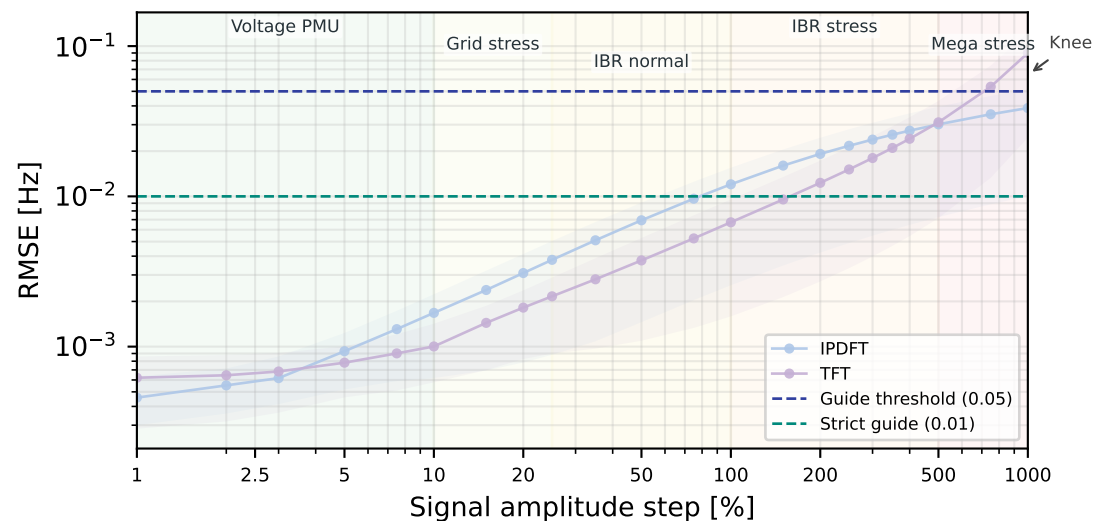
Loop-based



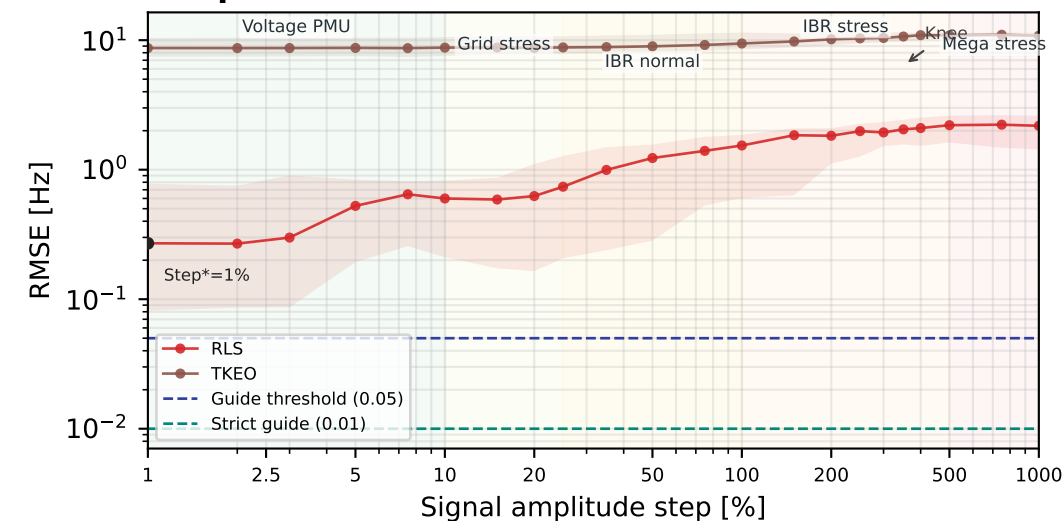
Model-based



Window-based



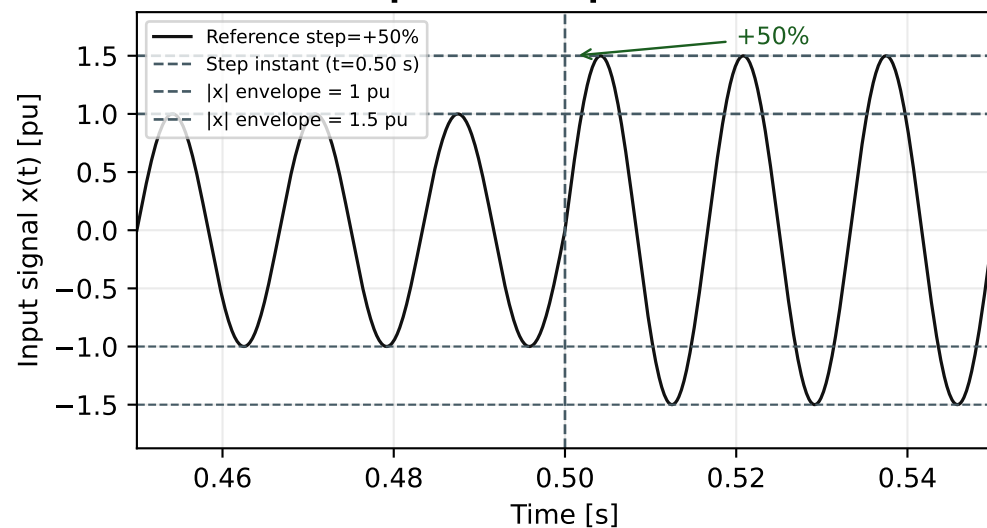
Adaptive



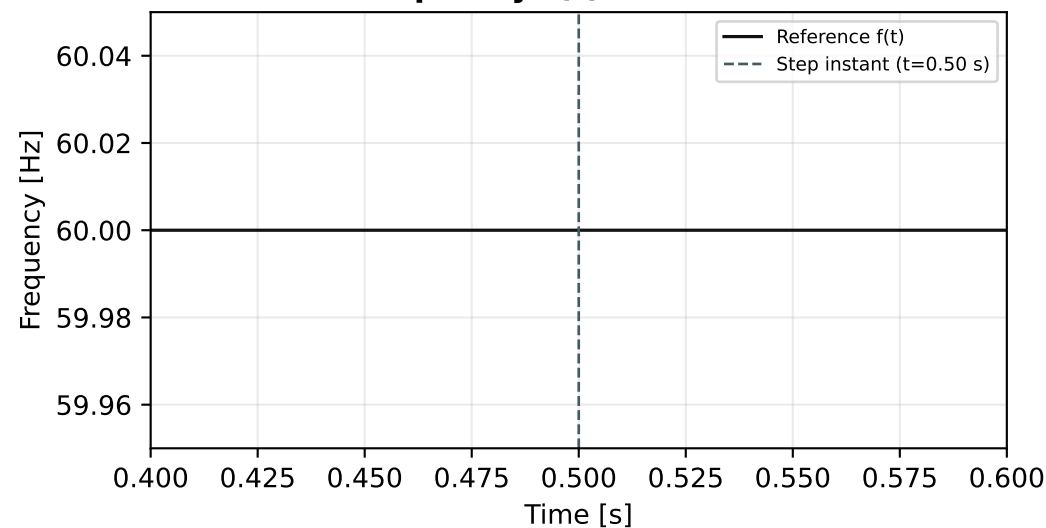
Post-step 1-cycle RMSE: by estimator family (Fixed Policy)

This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify magnitude-step cross-sensitivity, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits. Fixed-policy mode: one parameter set per estimator is reused across all amplitudes.

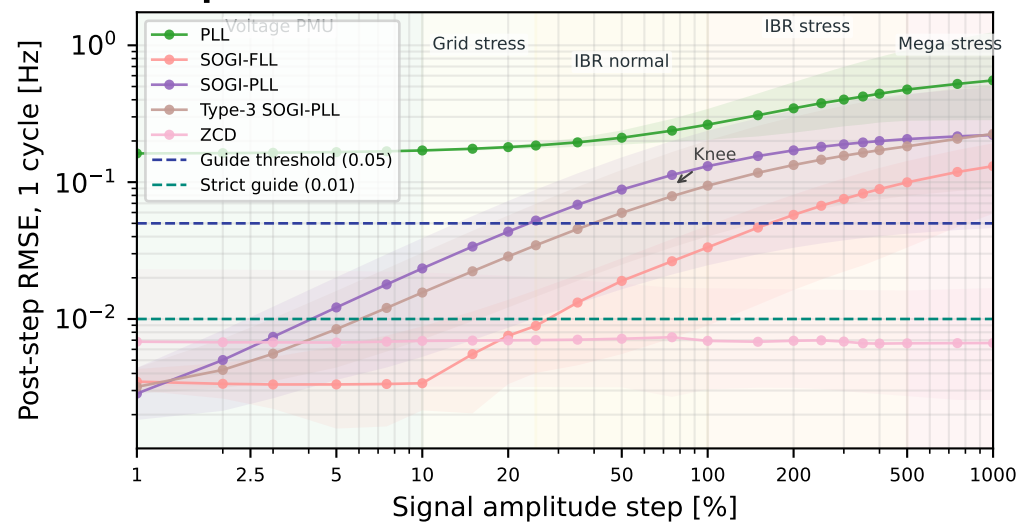
Reference Amplitude Step



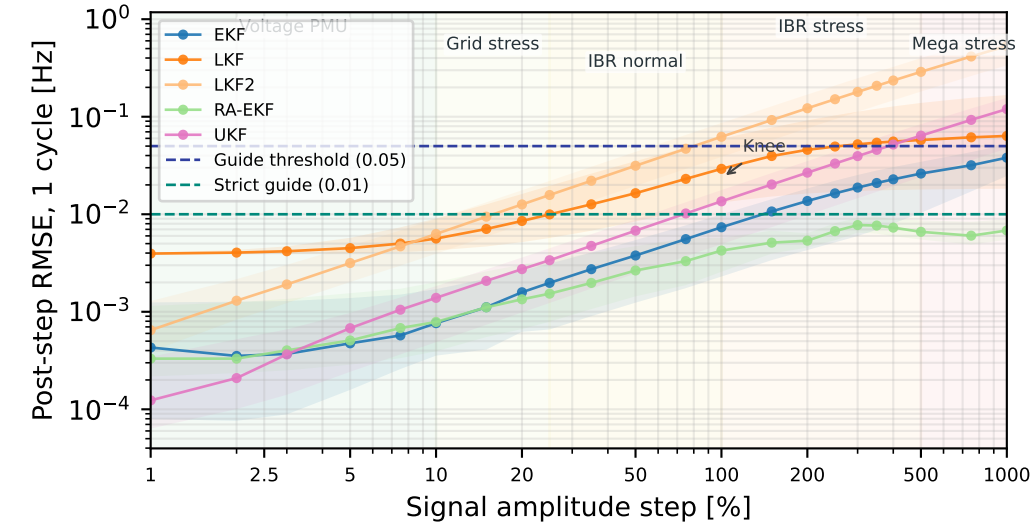
Reference Frequency f(t)



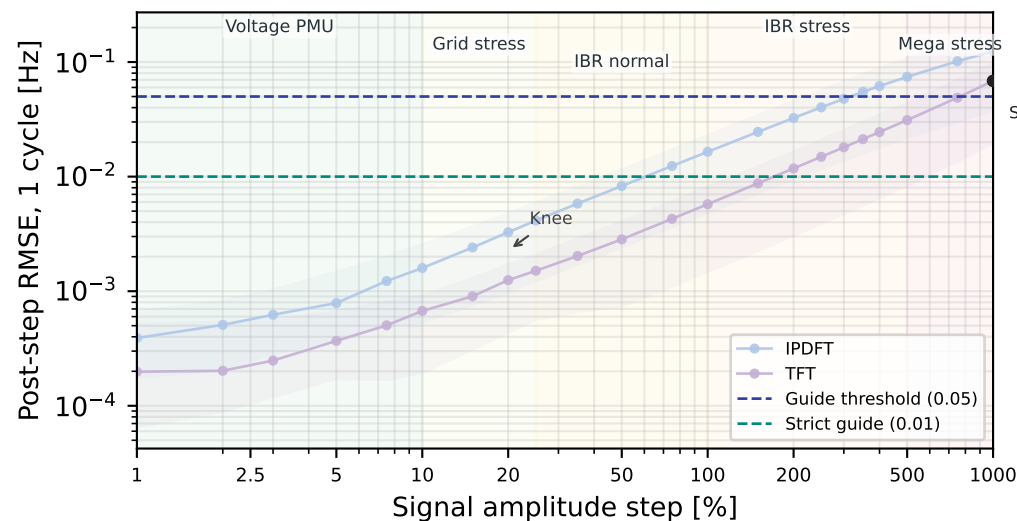
Loop-based



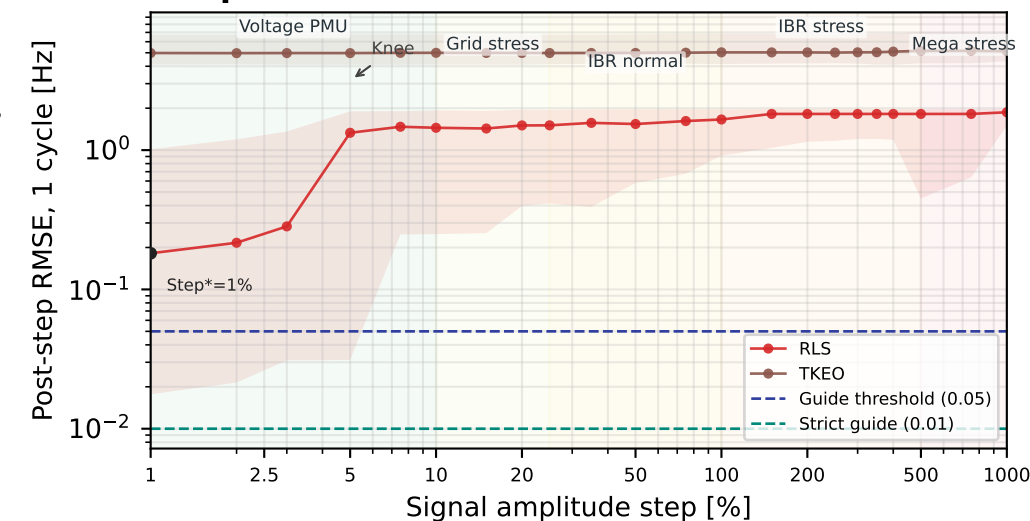
Model-based



Window-based



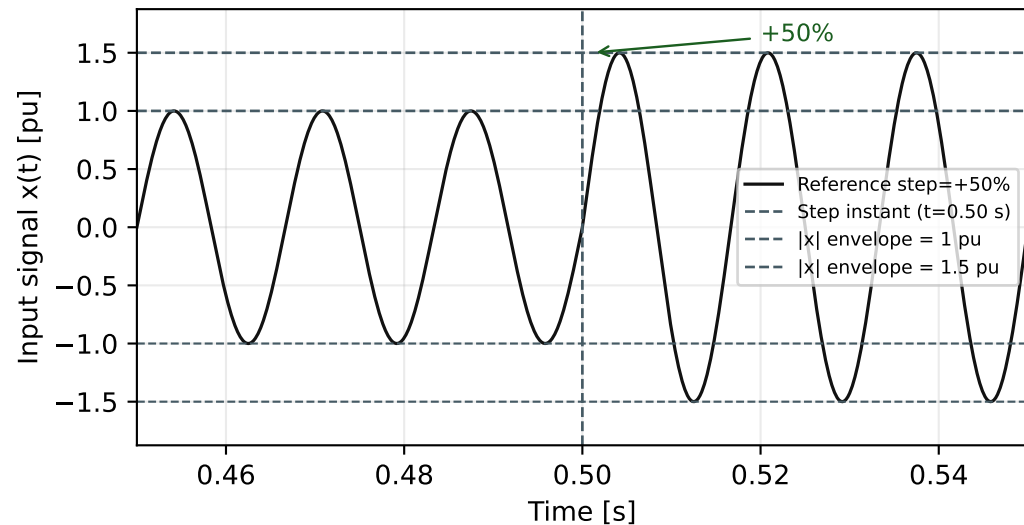
Adaptive



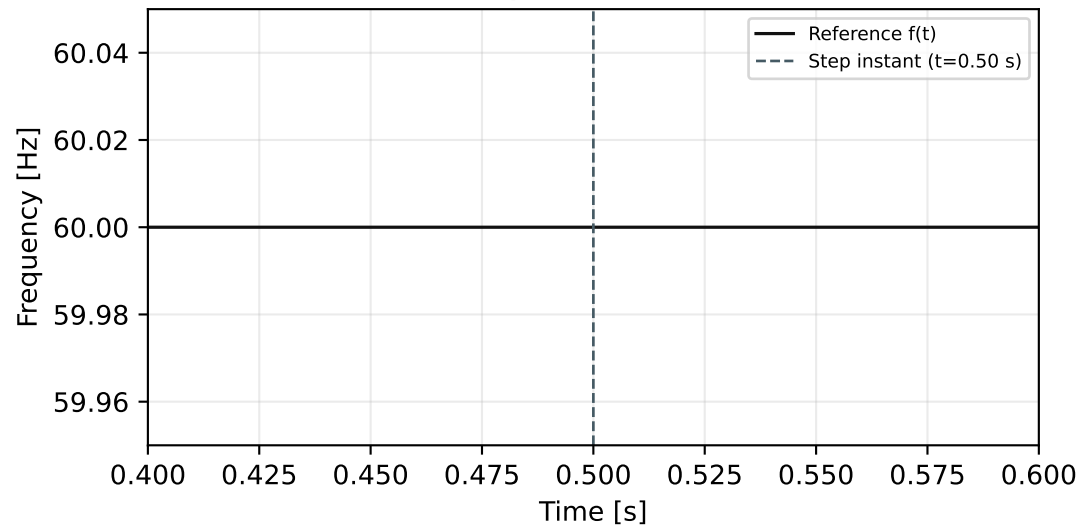
Post-step 100-ms RMSE: by estimator family (Fixed Policy)

This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify magnitude-step cross-sensitivity, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits. Fixed-policy mode: one parameter set per estimator is reused across all amplitudes.

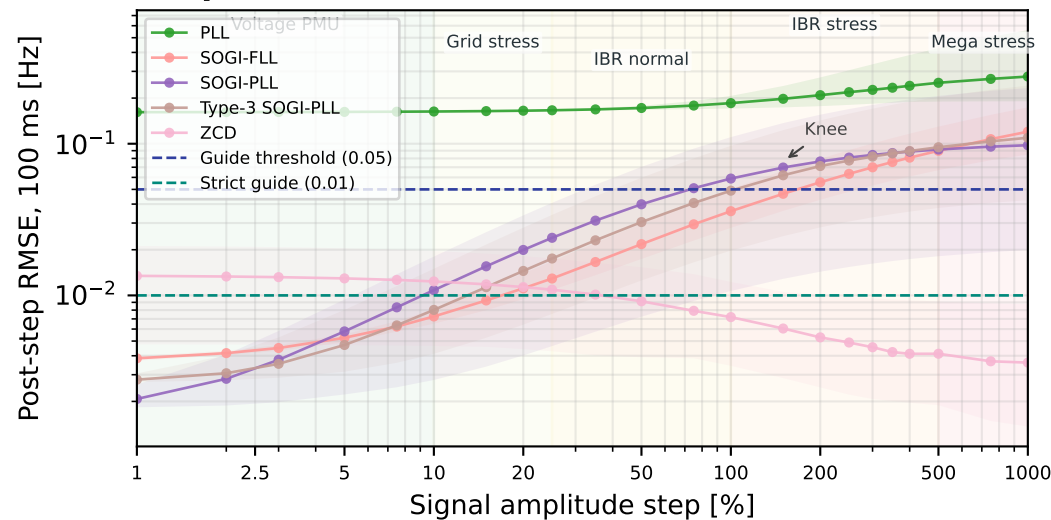
Reference Amplitude Step



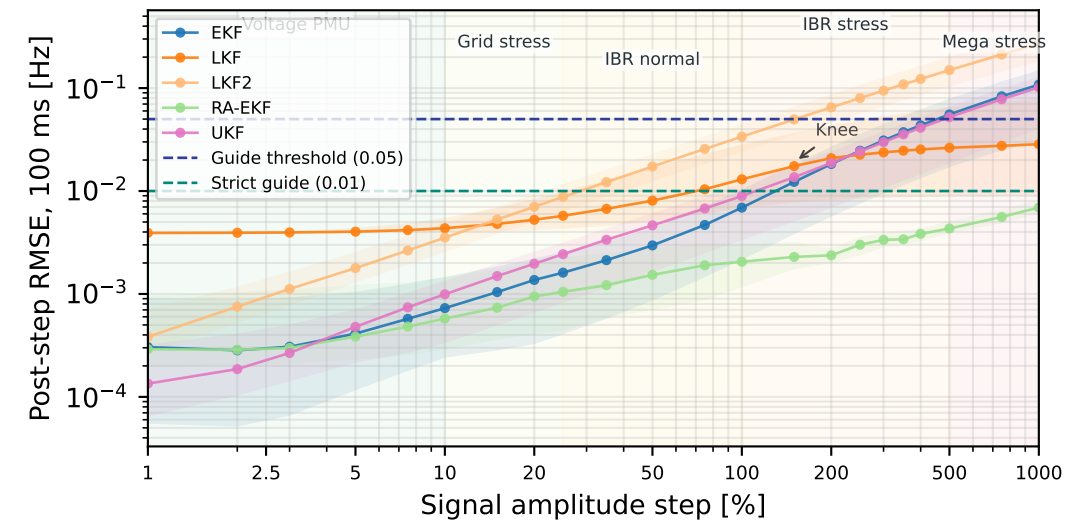
Reference Frequency $f(t)$



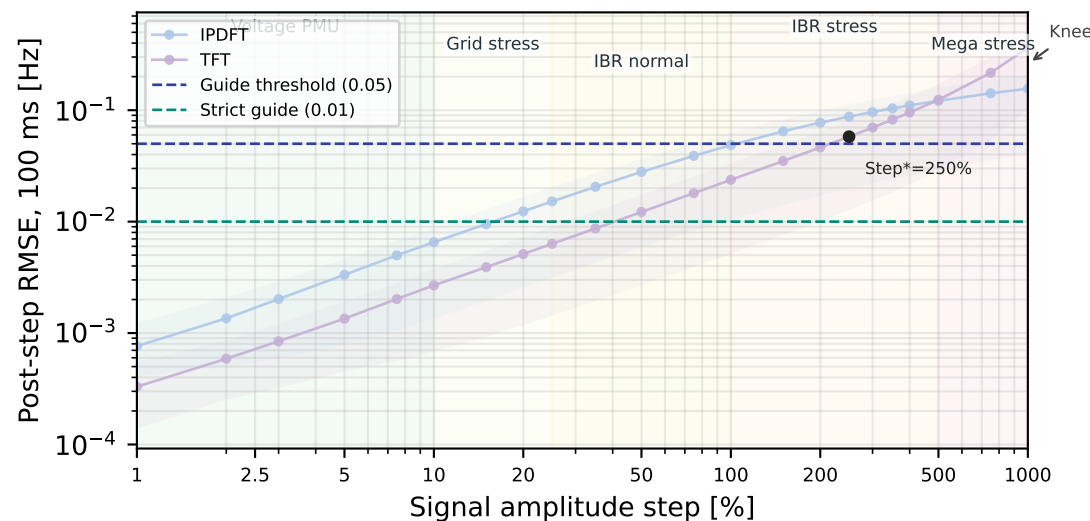
Loop-based



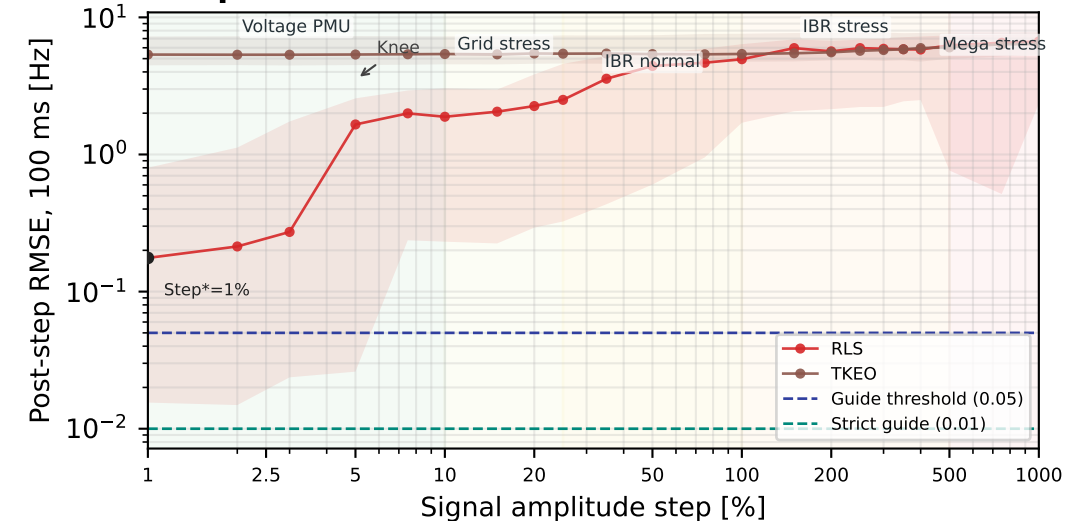
Model-based



Window-based



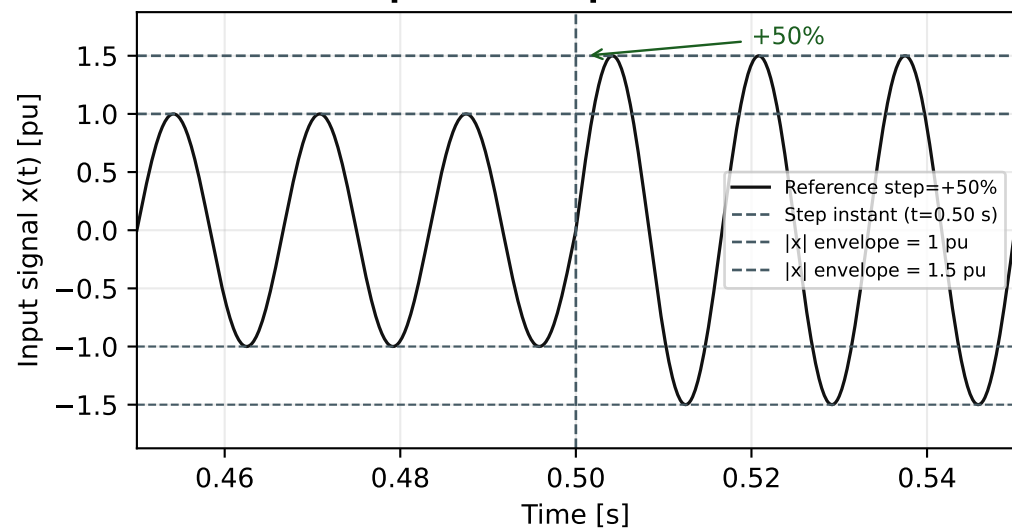
Adaptive



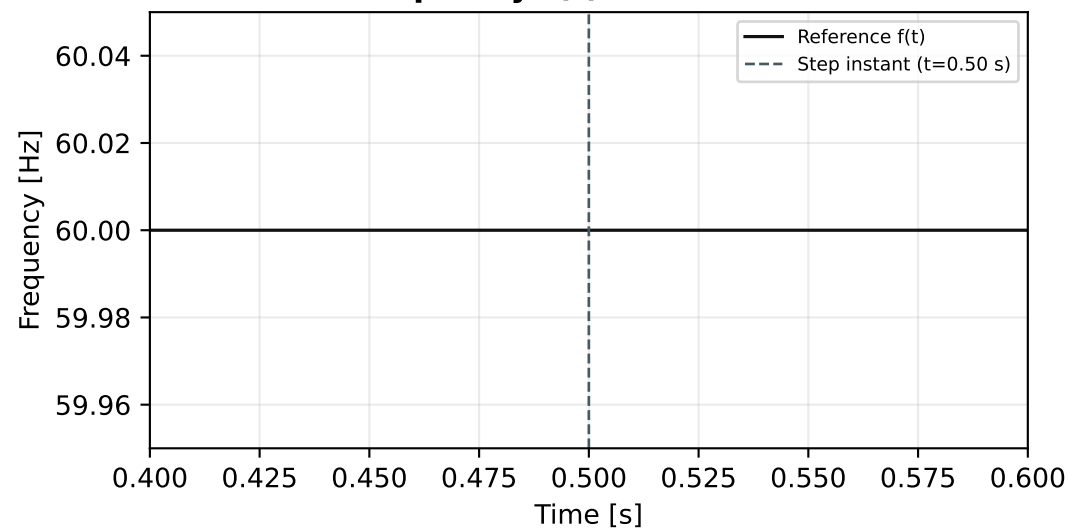
Post-step peak FE: by estimator family (Fixed Policy)

This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify magnitude-step cross-sensitivity, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits. Fixed-policy mode: one parameter set per estimator is reused across all amplitudes.

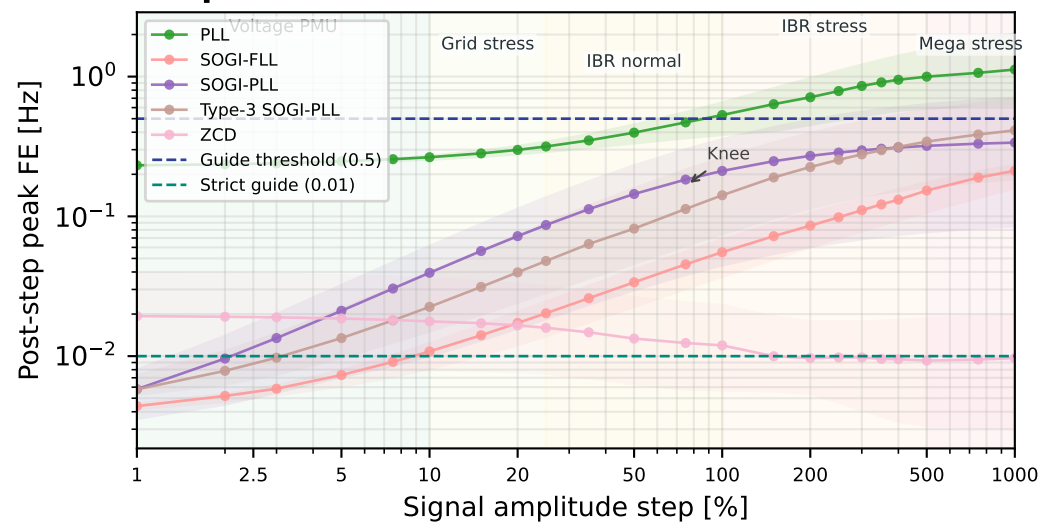
Reference Amplitude Step



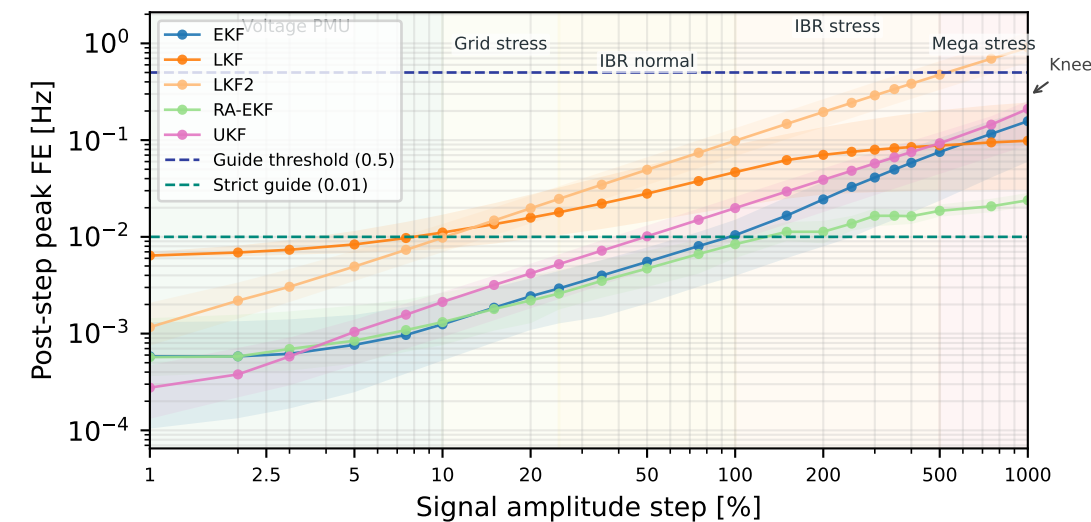
Reference Frequency $f(t)$



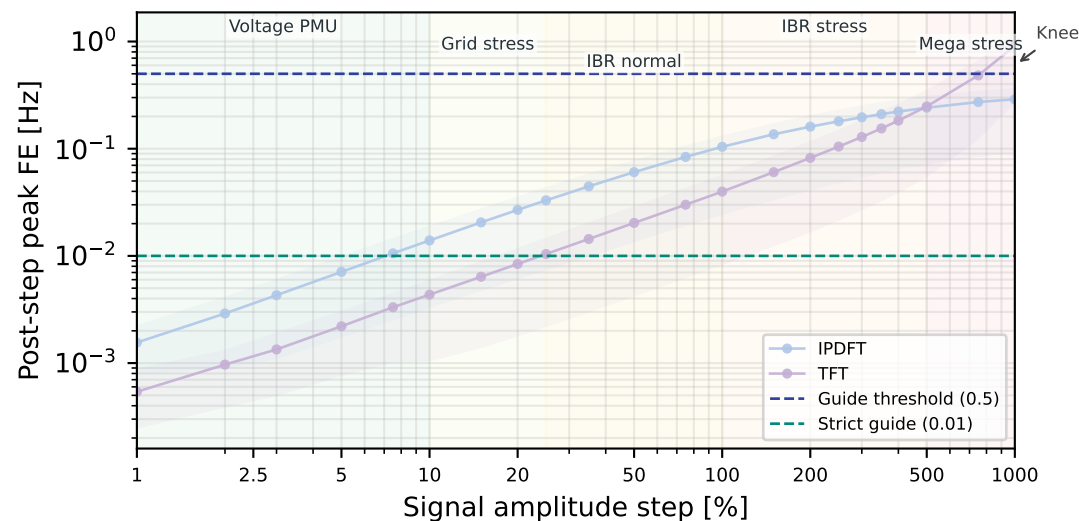
Loop-based



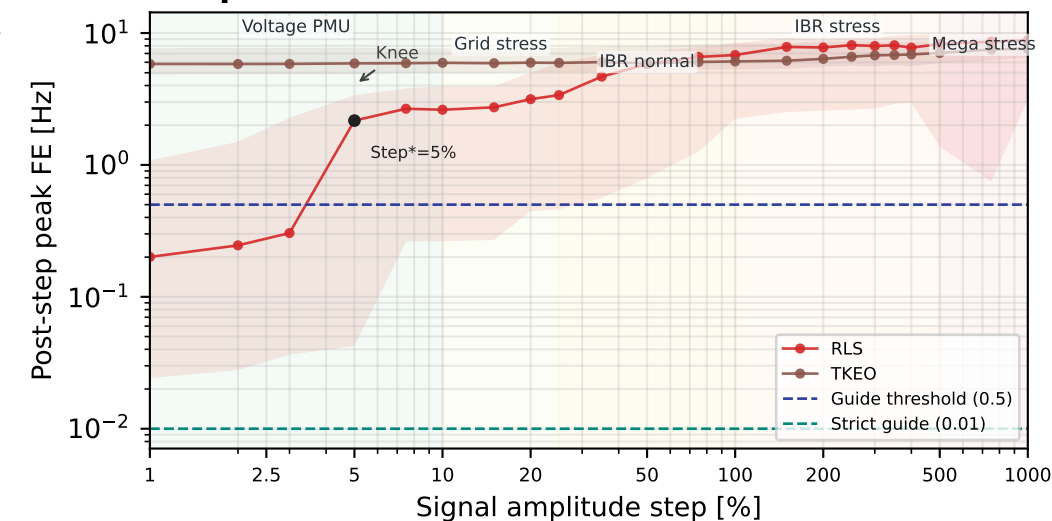
Model-based



Window-based



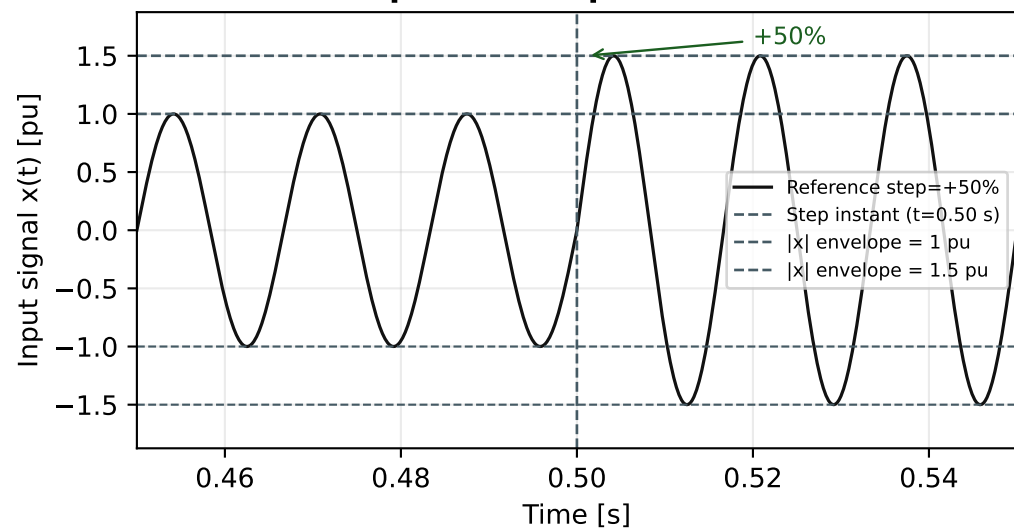
Adaptive



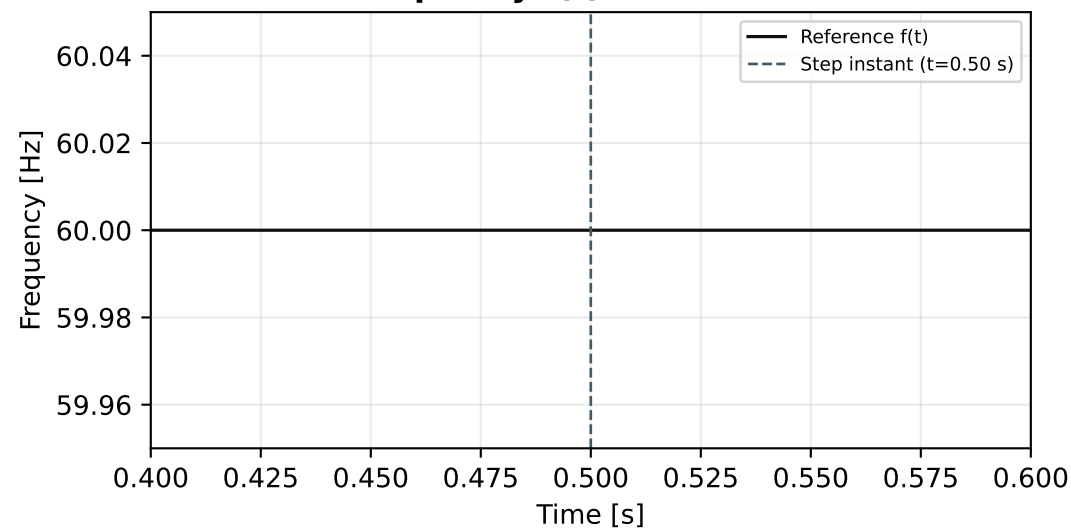
Event settling time: by estimator family (Fixed Policy)

This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify magnitude-step cross-sensitivity, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits. Fixed-policy mode: one parameter set per estimator is reused across all amplitudes.

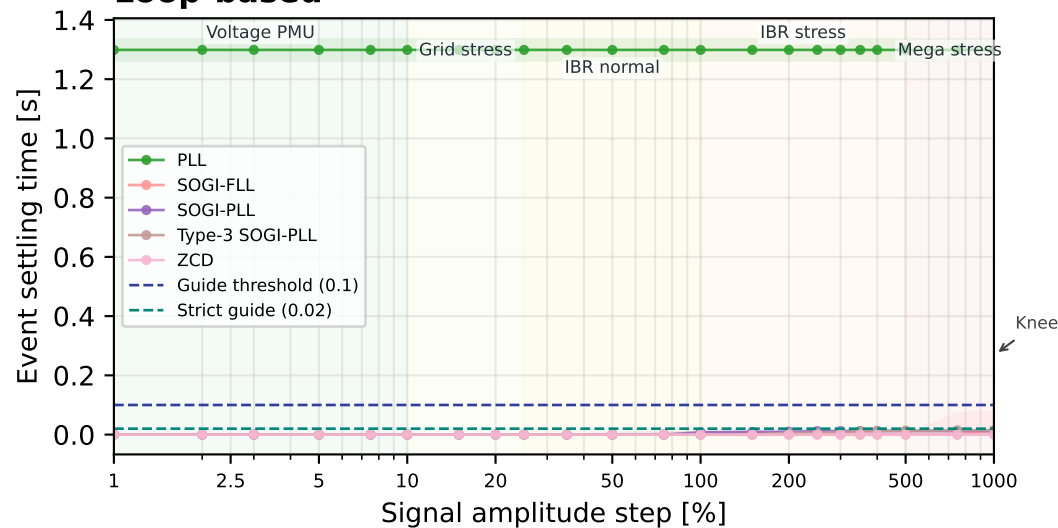
Reference Amplitude Step



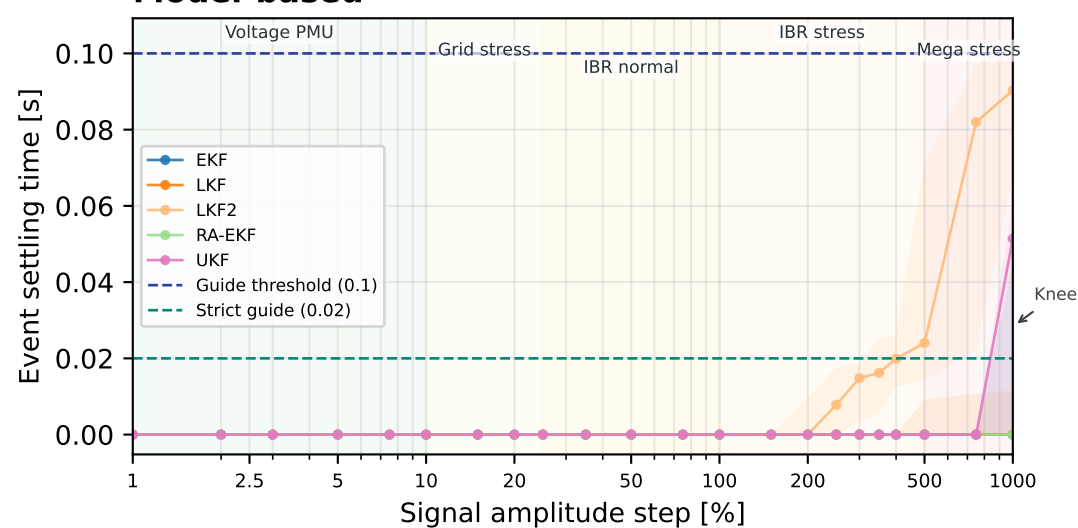
Reference Frequency $f(t)$



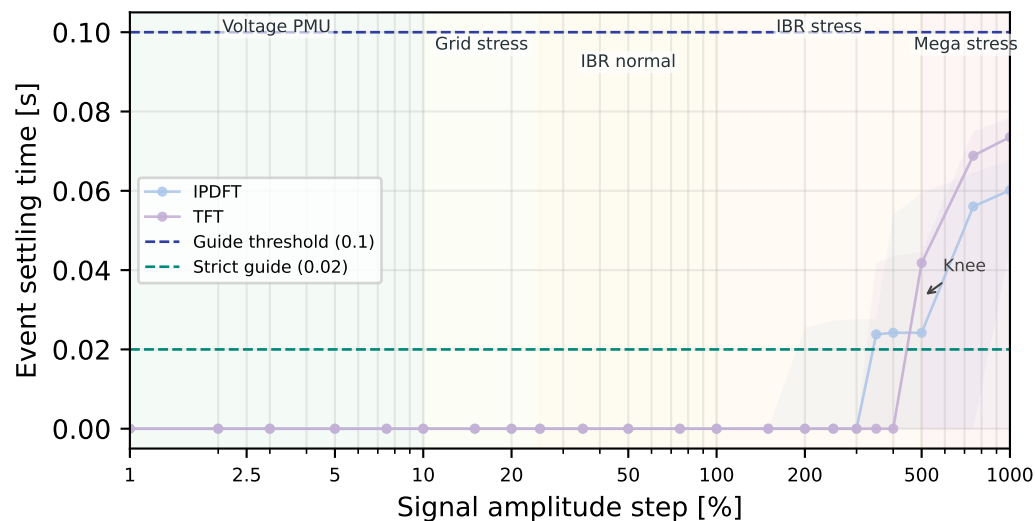
Loop-based



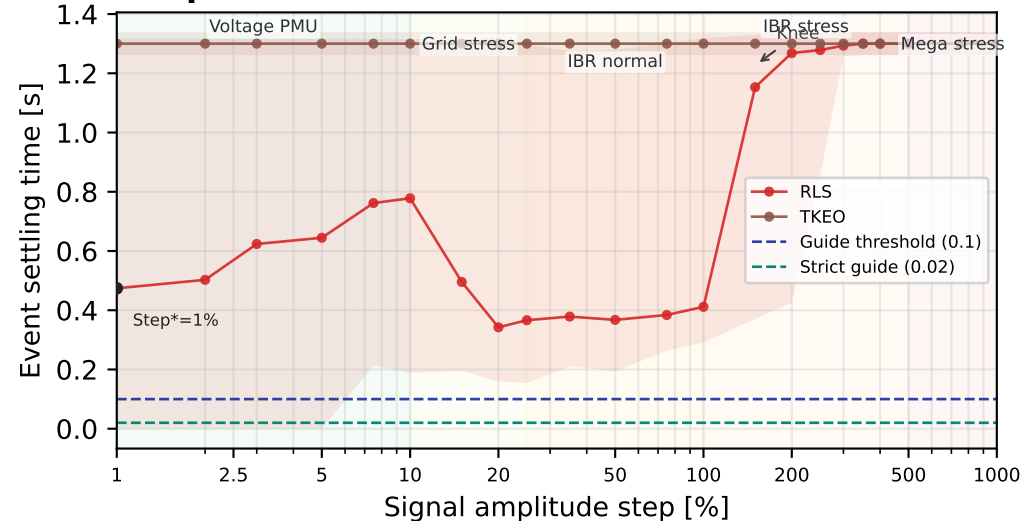
Model-based



Window-based



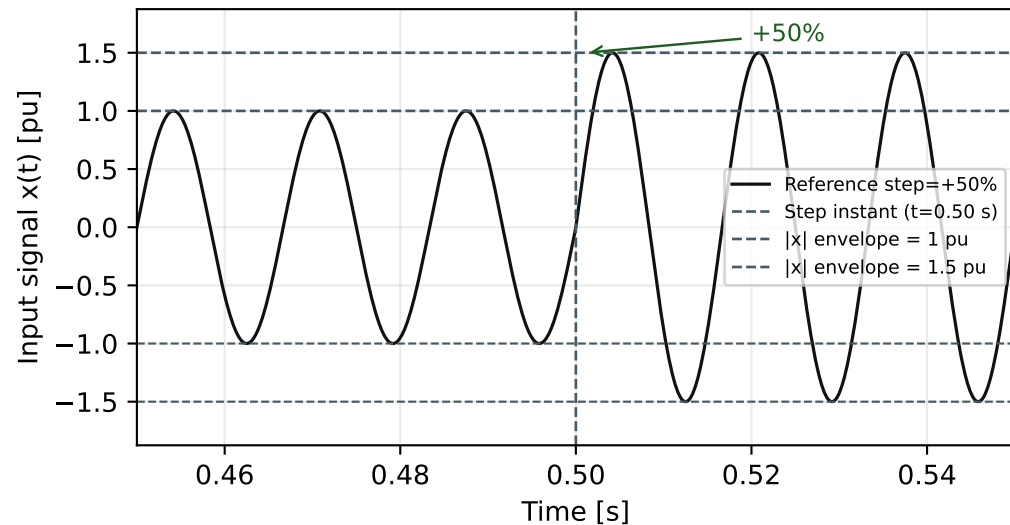
Adaptive



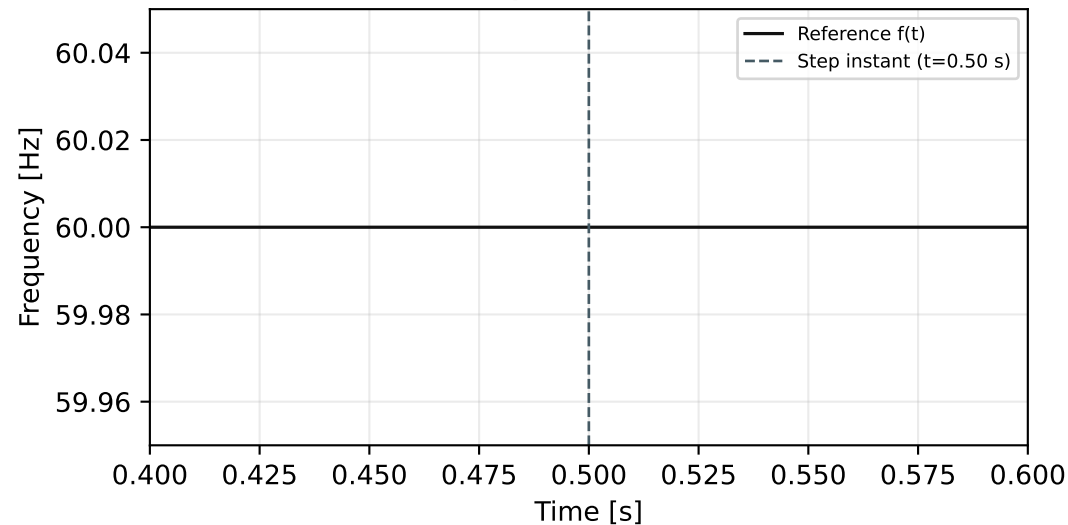
FE max per test: by estimator family (Fixed Policy)

This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify magnitude-step cross-sensitivity, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits. Fixed-policy mode: one parameter set per estimator is reused across all amplitudes.

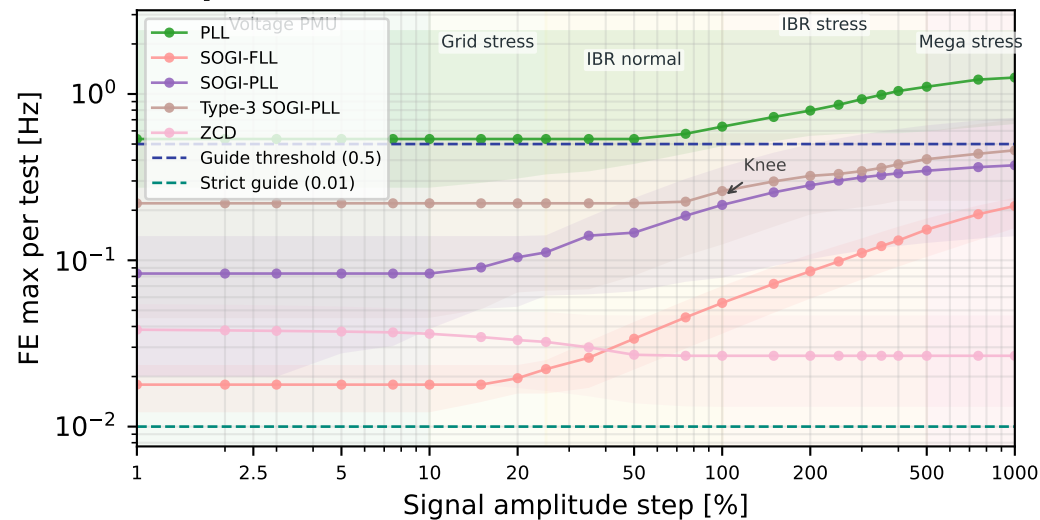
Reference Amplitude Step



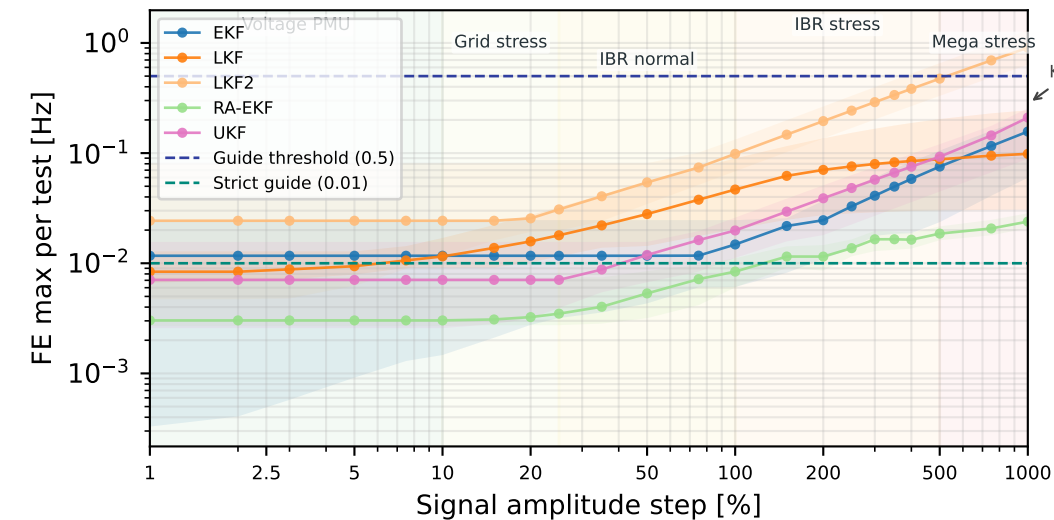
Reference Frequency f(t)



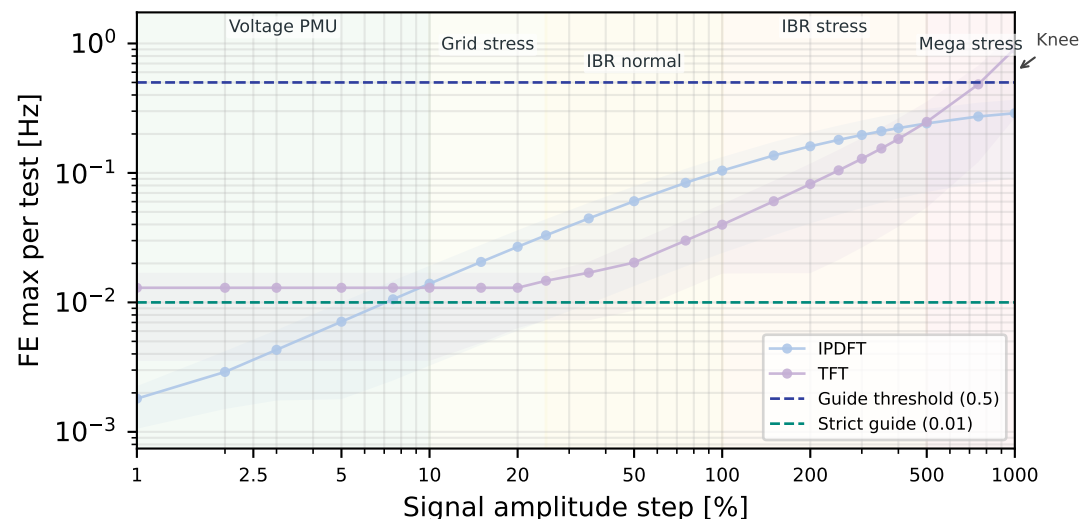
Loop-based



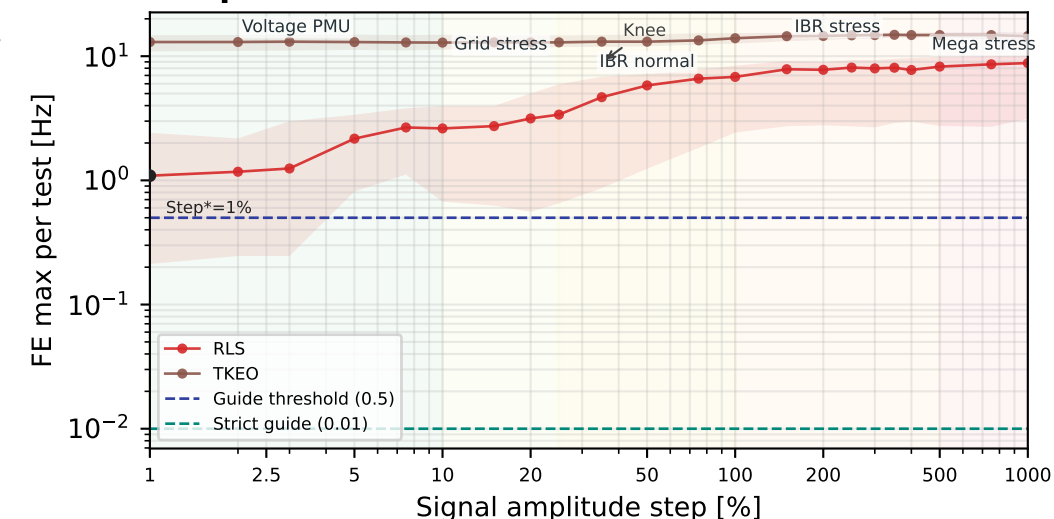
Model-based



Window-based



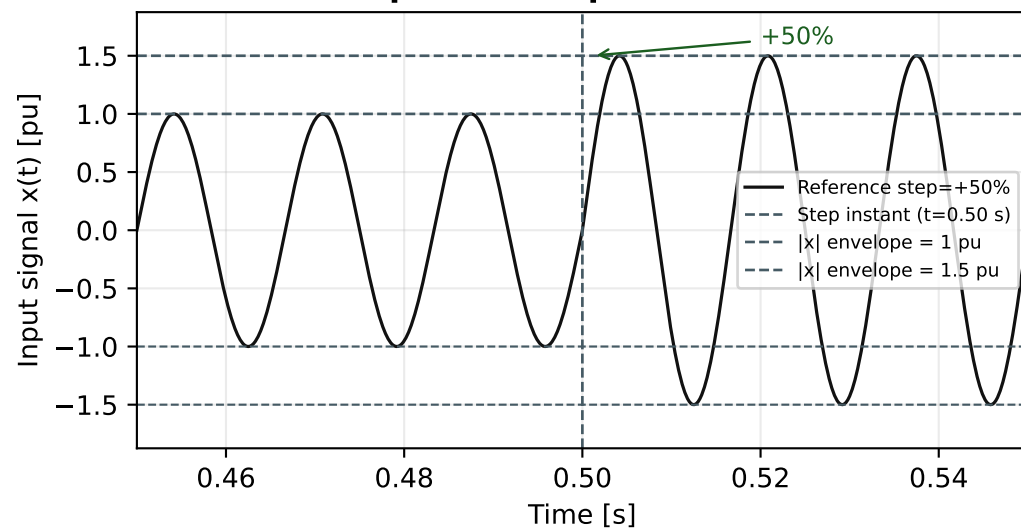
Adaptive



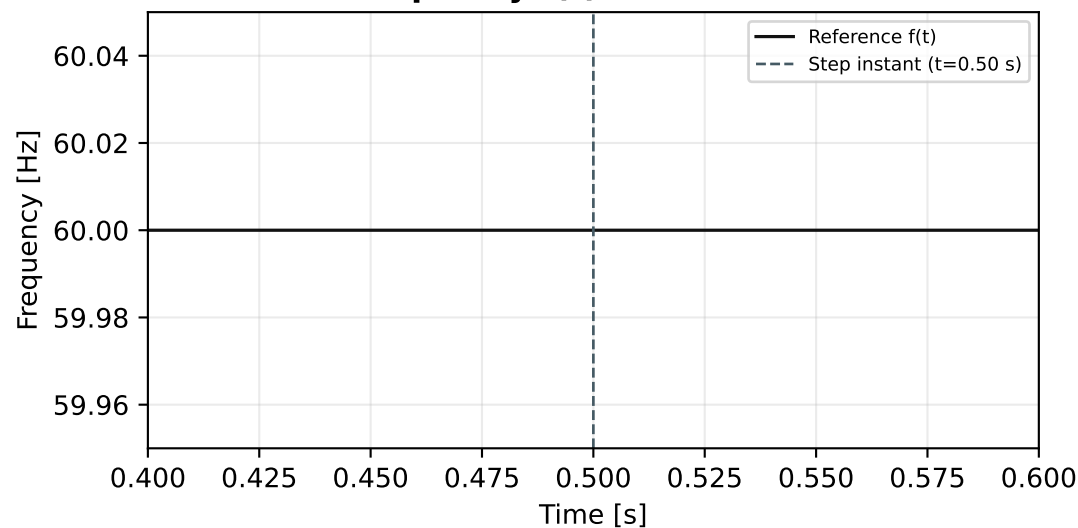
RFE: by estimator family (Fixed Policy)

This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify magnitude-step cross-sensitivity, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits. Fixed-policy mode: one parameter set per estimator is reused across all amplitudes.

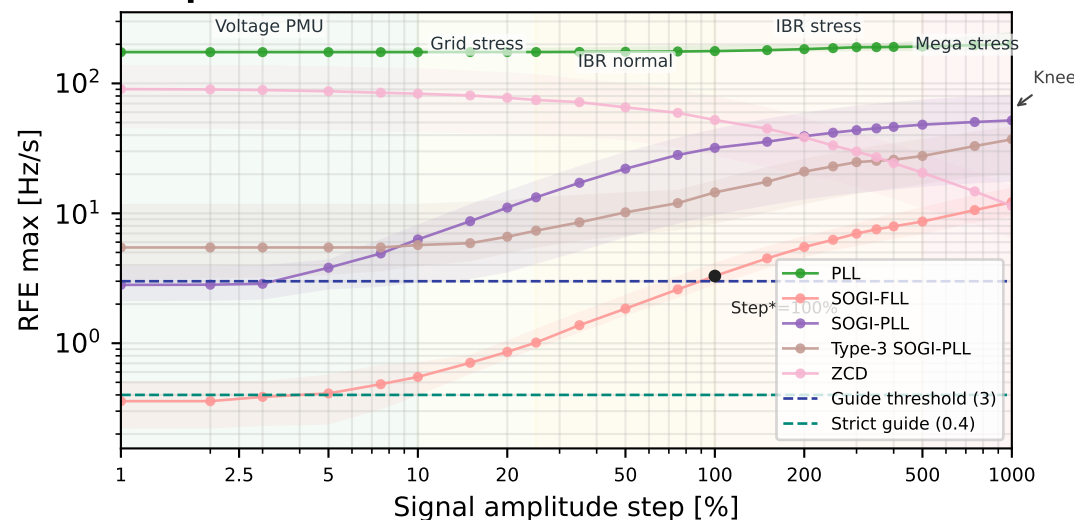
Reference Amplitude Step



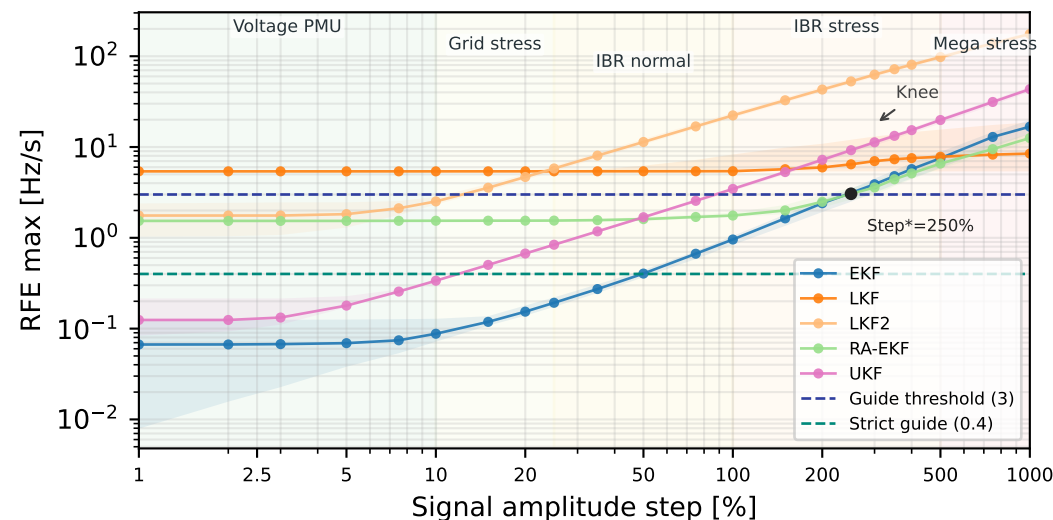
Reference Frequency $f(t)$



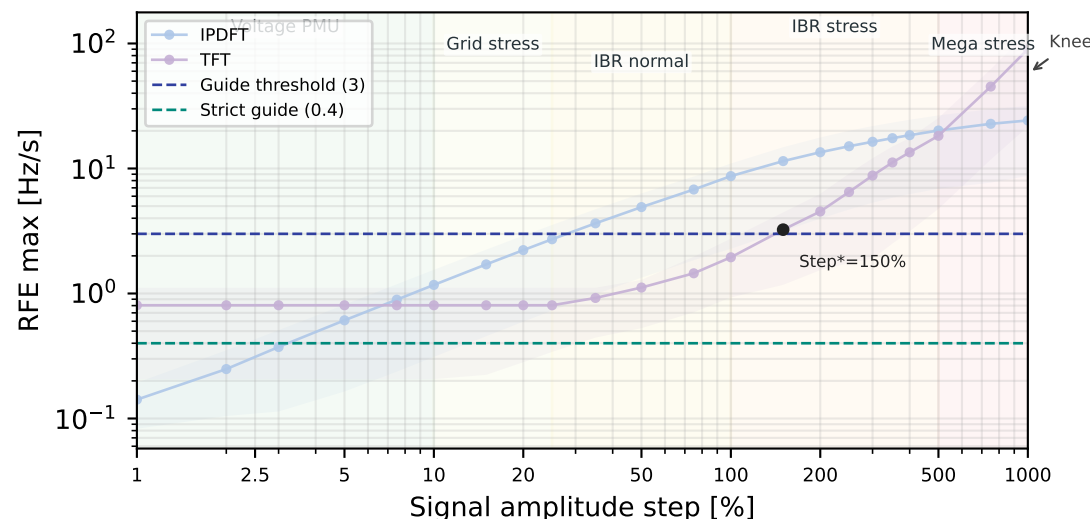
Loop-based



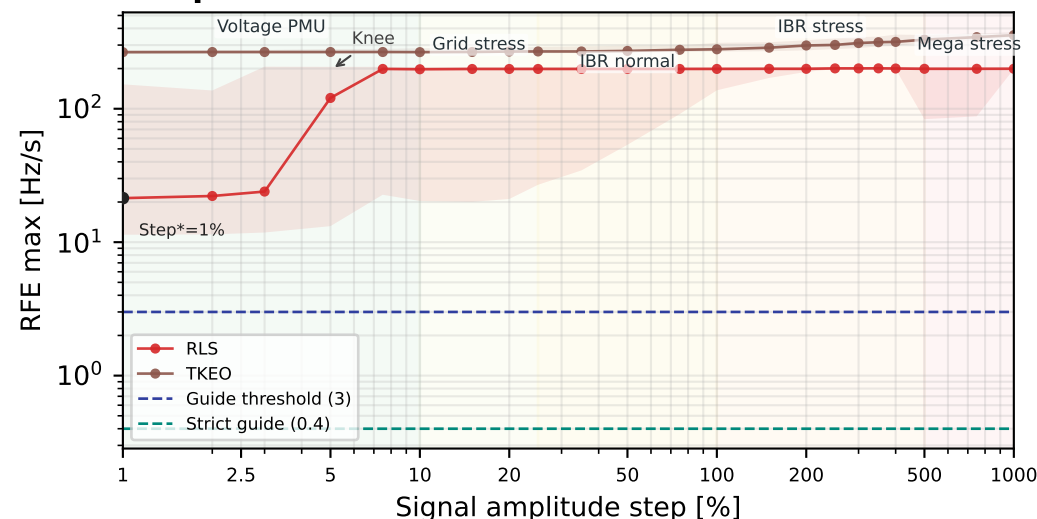
Model-based



Window-based

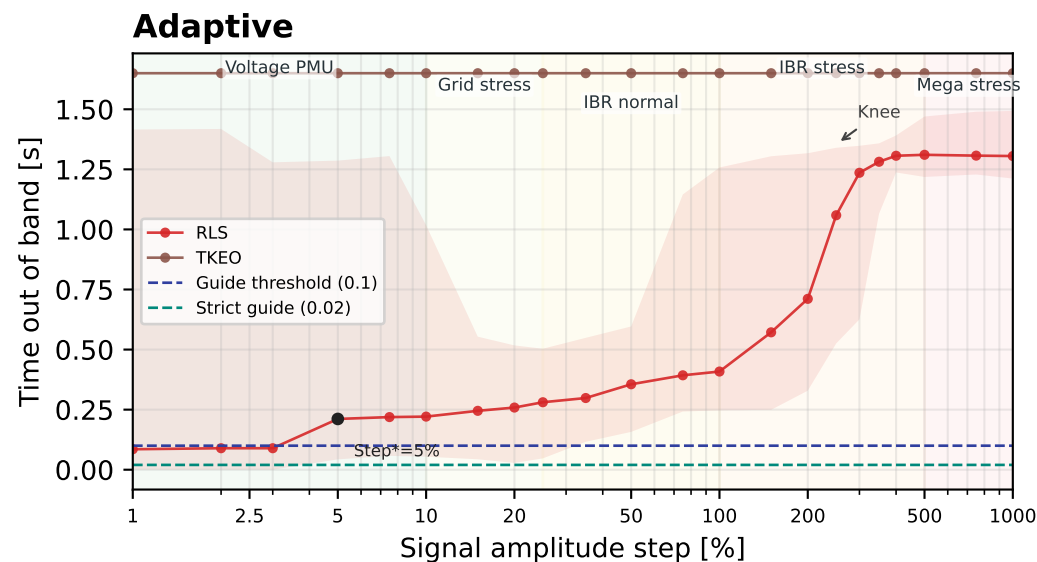
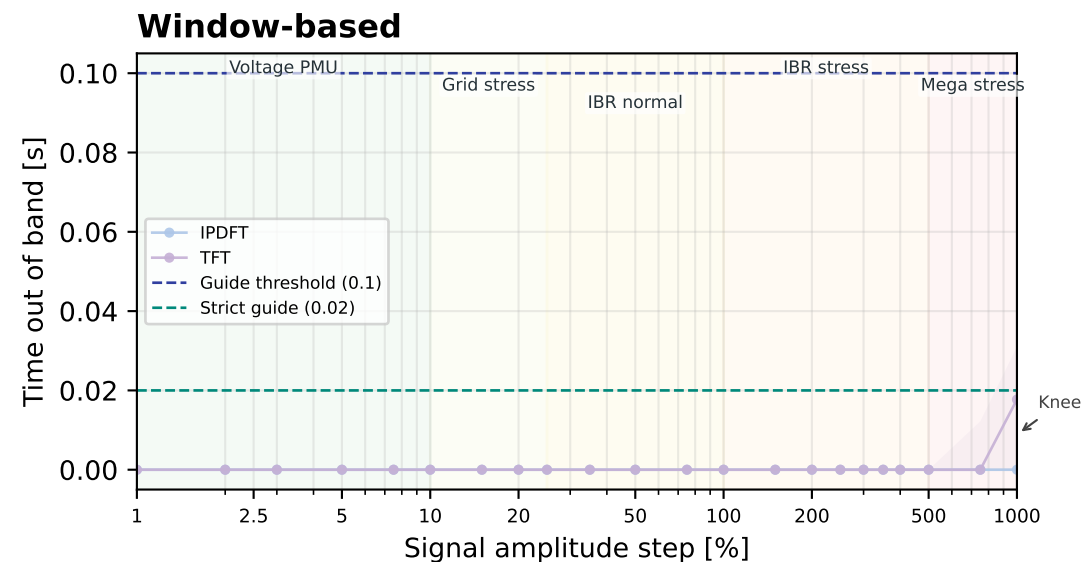
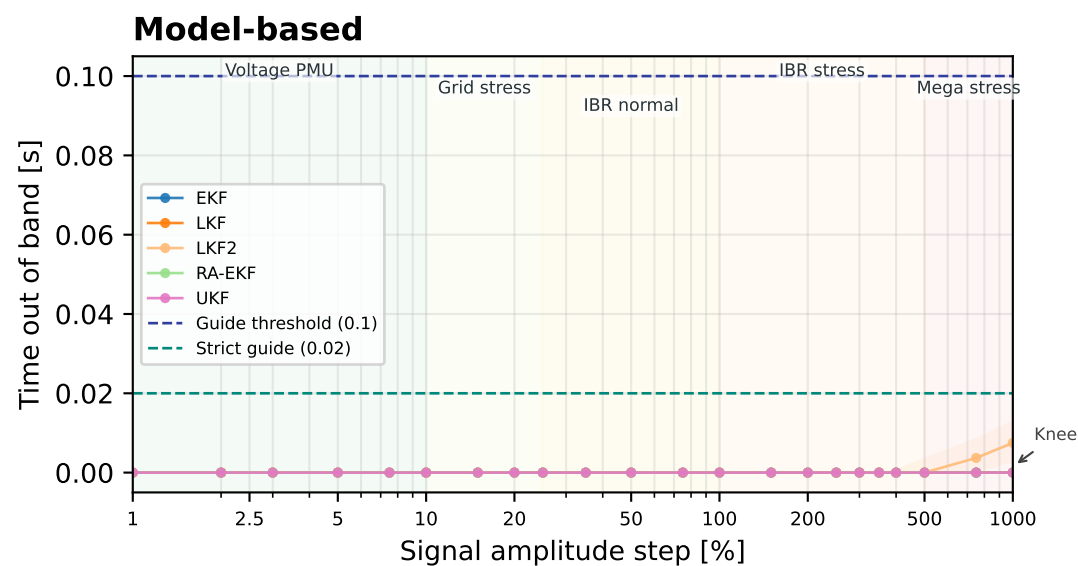
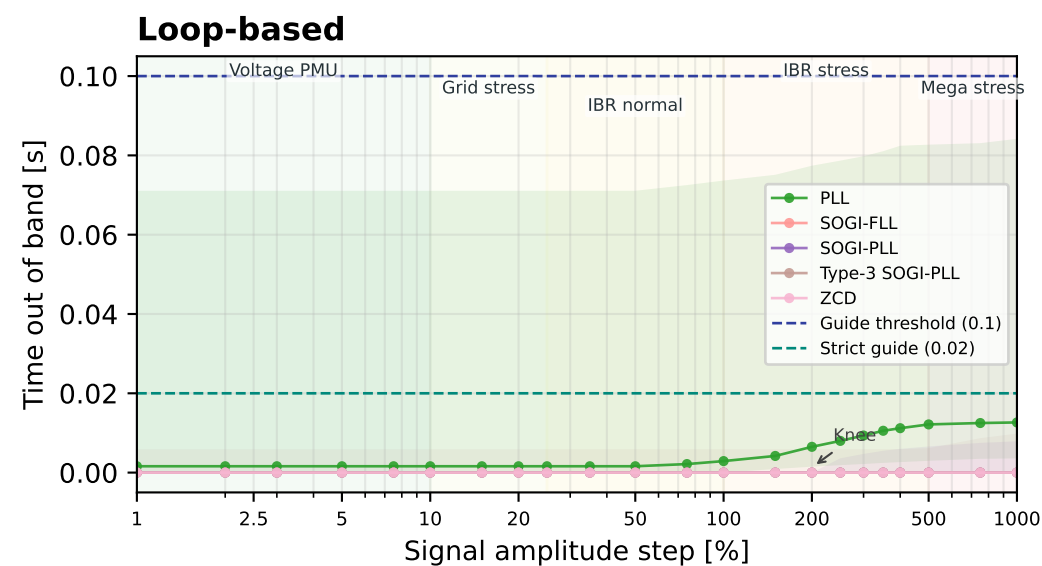
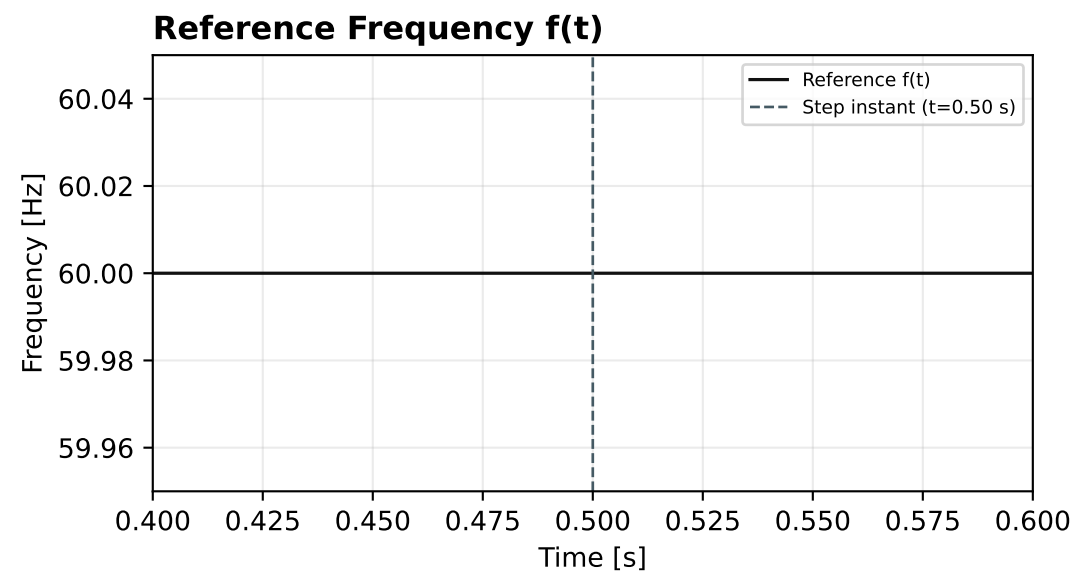
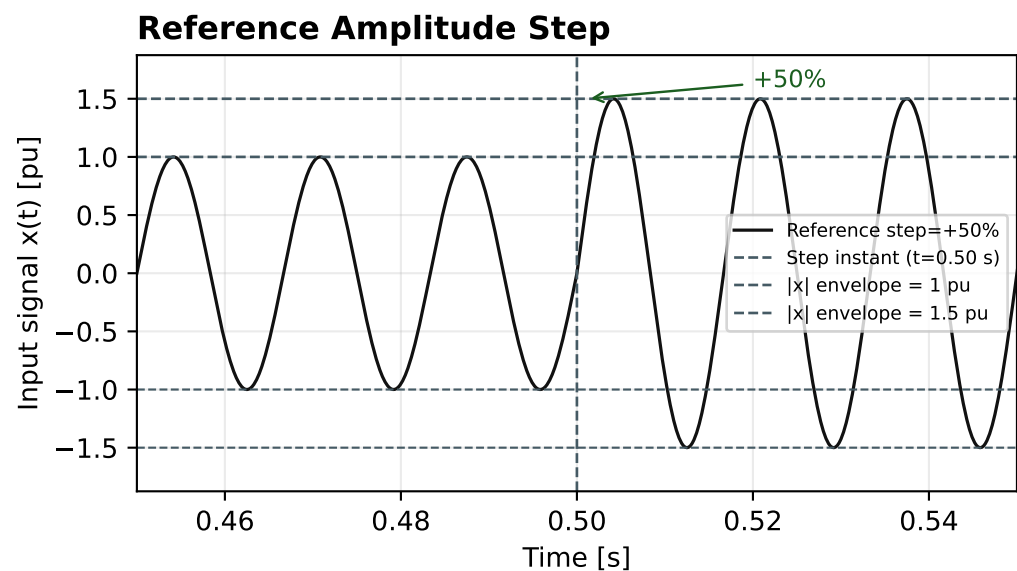


Adaptive



Time out of band: by estimator family (Fixed Policy)

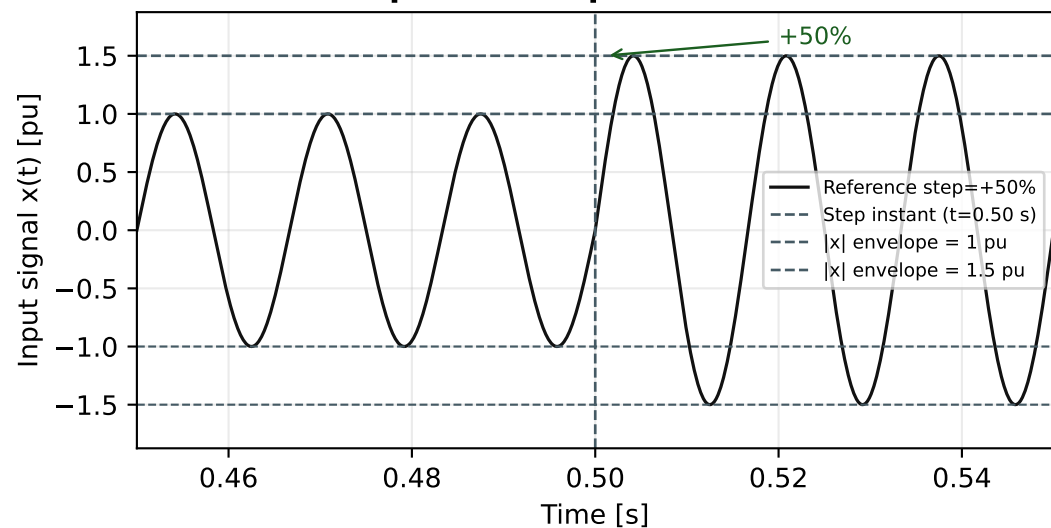
This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify magnitude-step cross-sensitivity, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits. Fixed-policy mode: one parameter set per estimator is reused across all amplitudes.



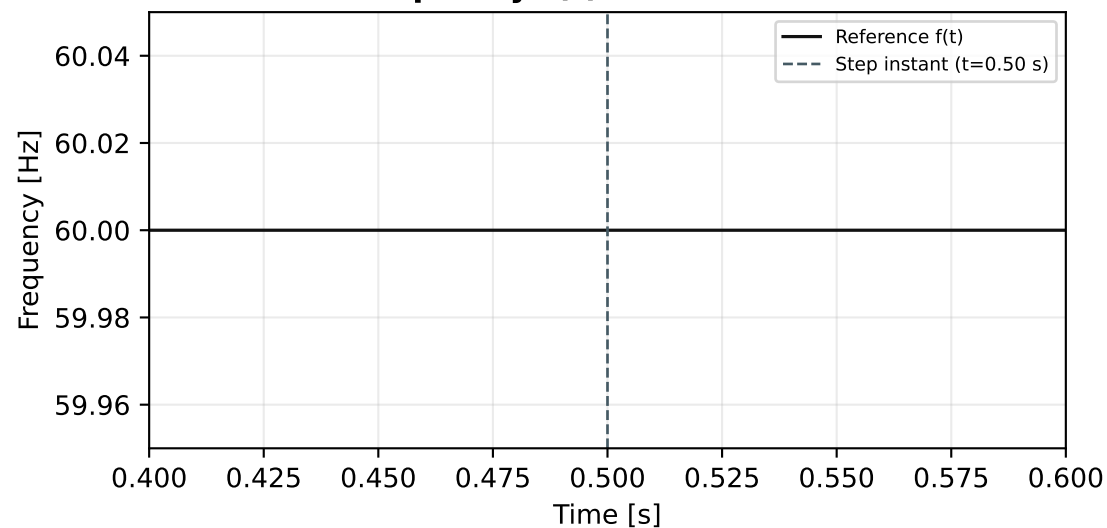
Pass rate: by estimator family (Fixed Policy)

This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify magnitude-step cross-sensitivity, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits. Fixed-policy mode: one parameter set per estimator is reused across all amplitudes.

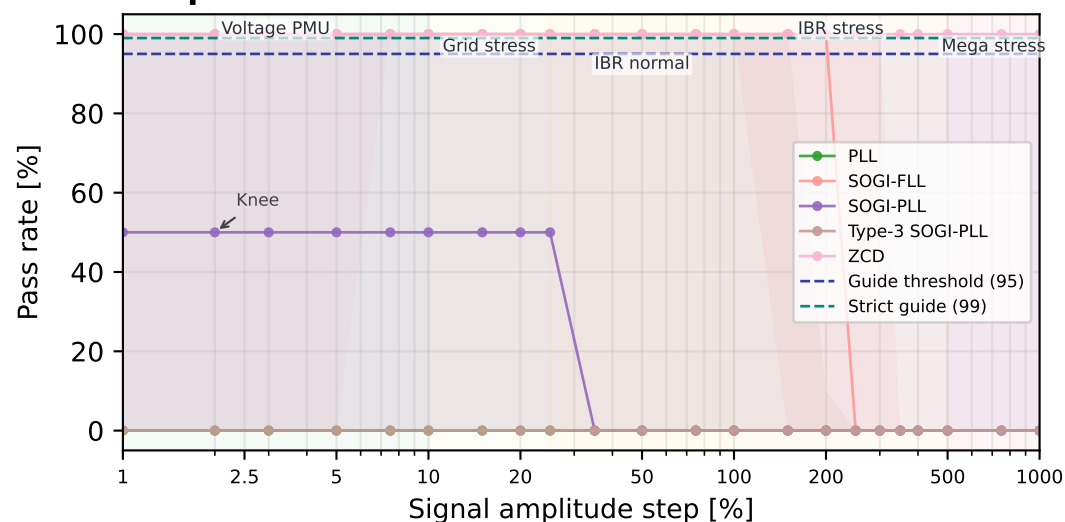
Reference Amplitude Step



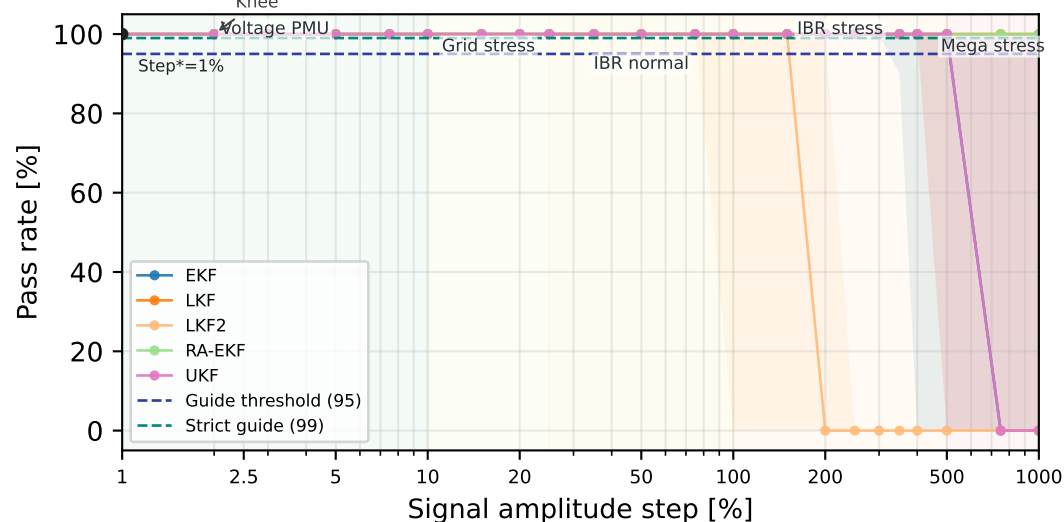
Reference Frequency $f(t)$



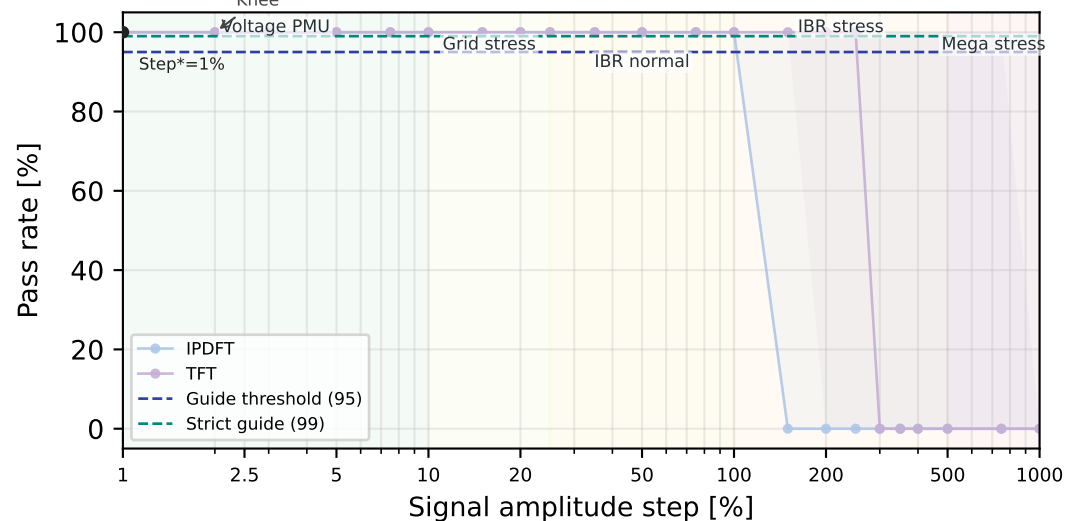
Loop-based



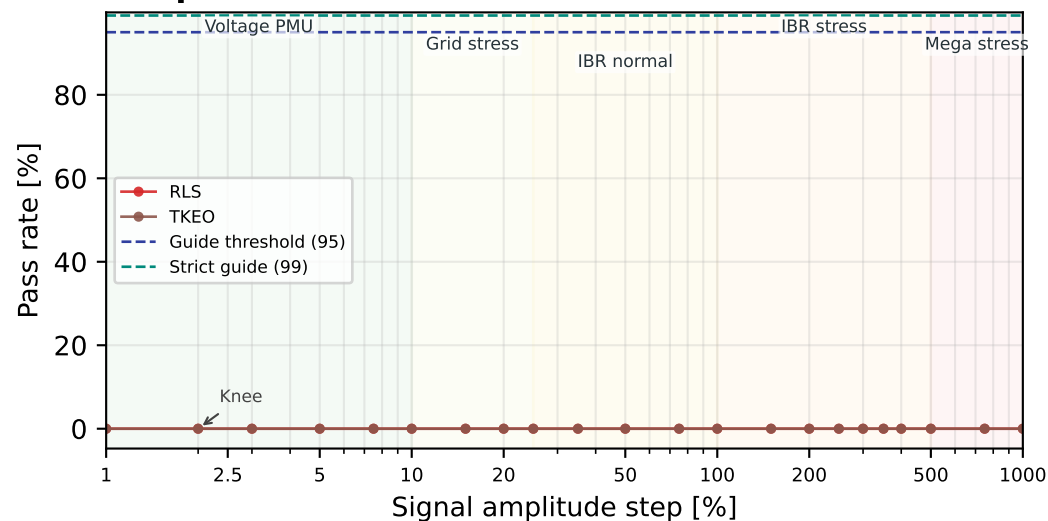
Model-based



Window-based



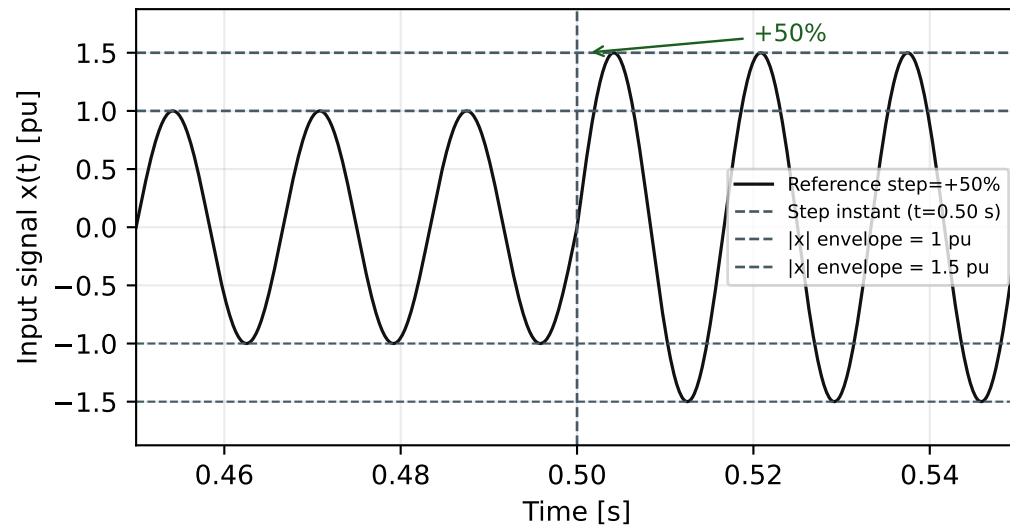
Adaptive



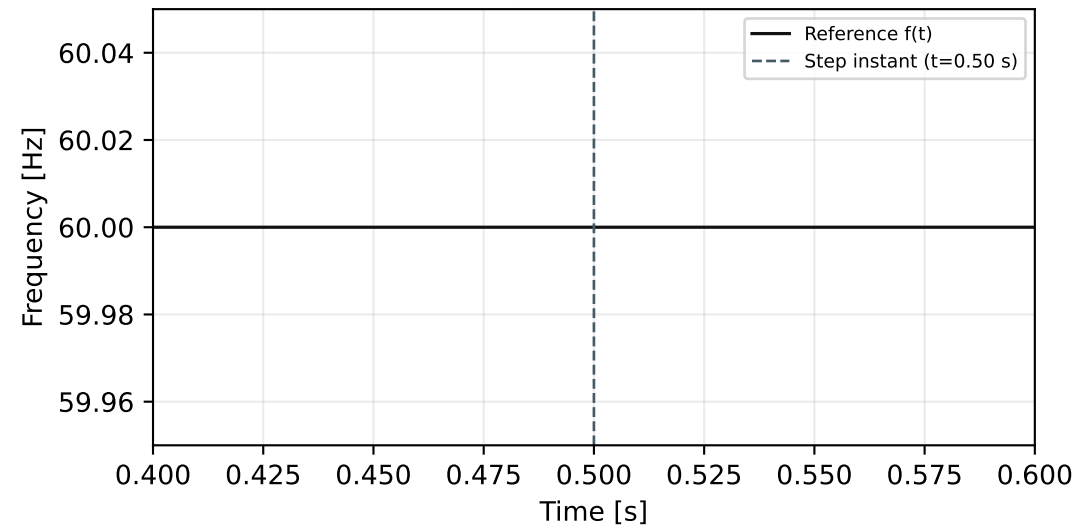
CPU cost: by estimator family (Fixed Policy)

This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify magnitude-step cross-sensitivity, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits. Fixed-policy mode: one parameter set per estimator is reused across all amplitudes.

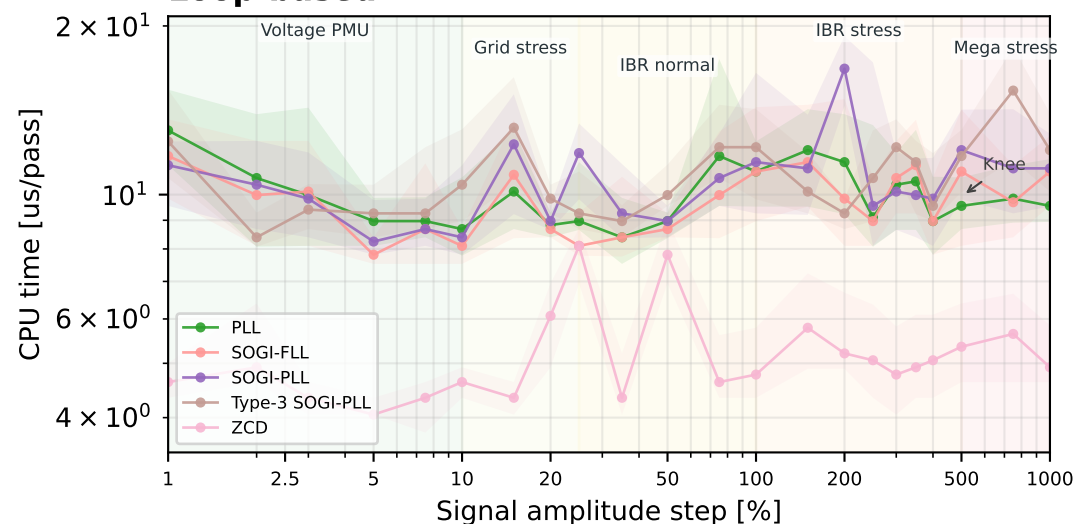
Reference Amplitude Step



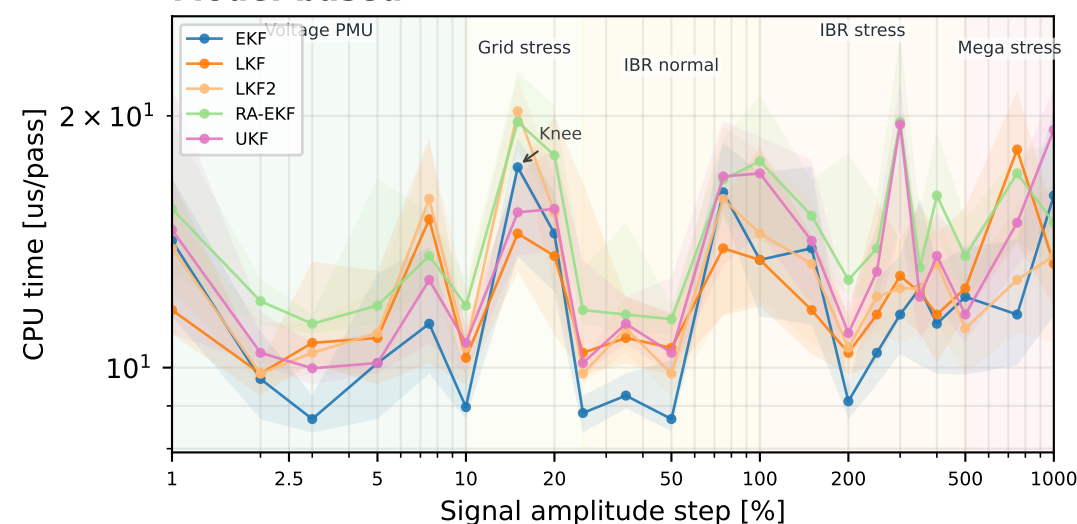
Reference Frequency f(t)



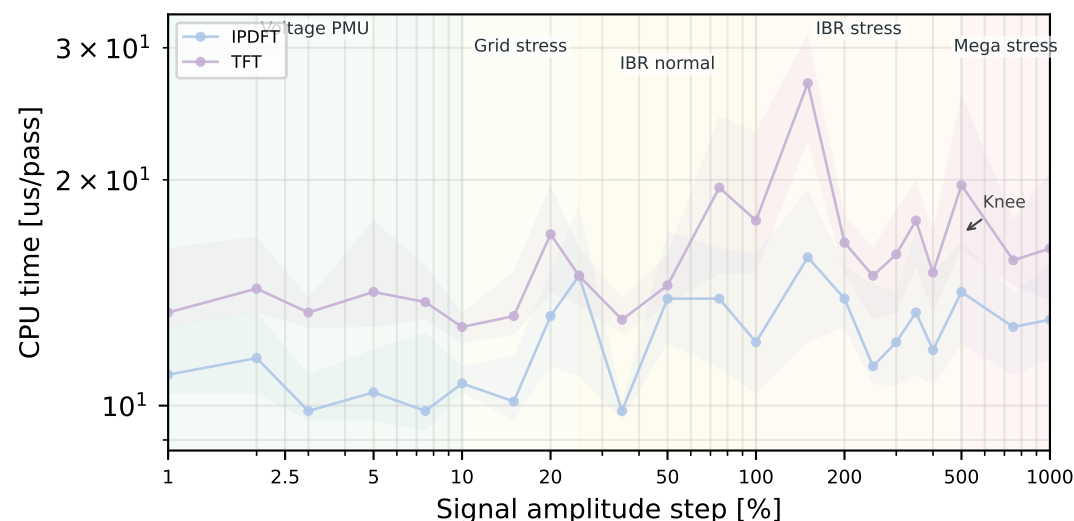
Loop-based



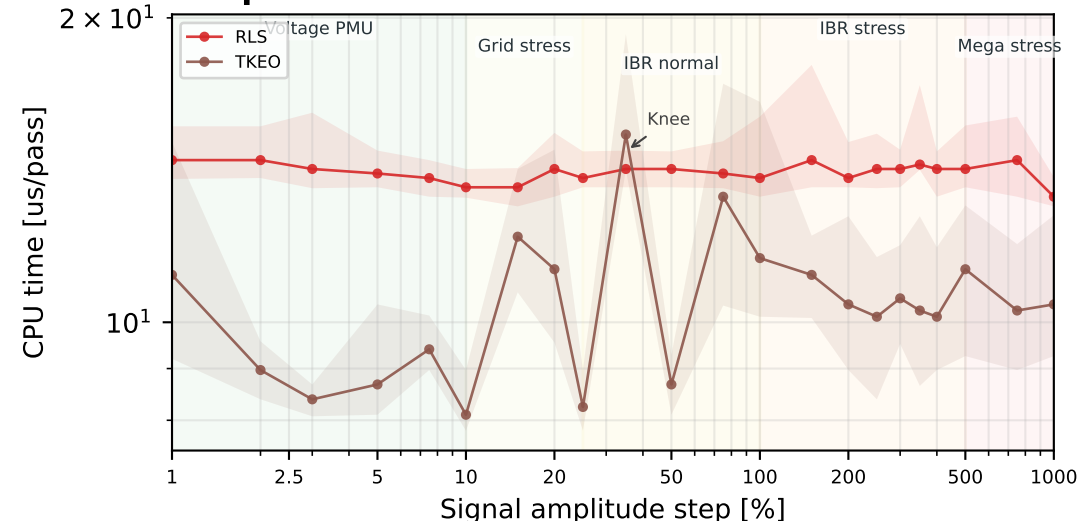
Model-based



Window-based



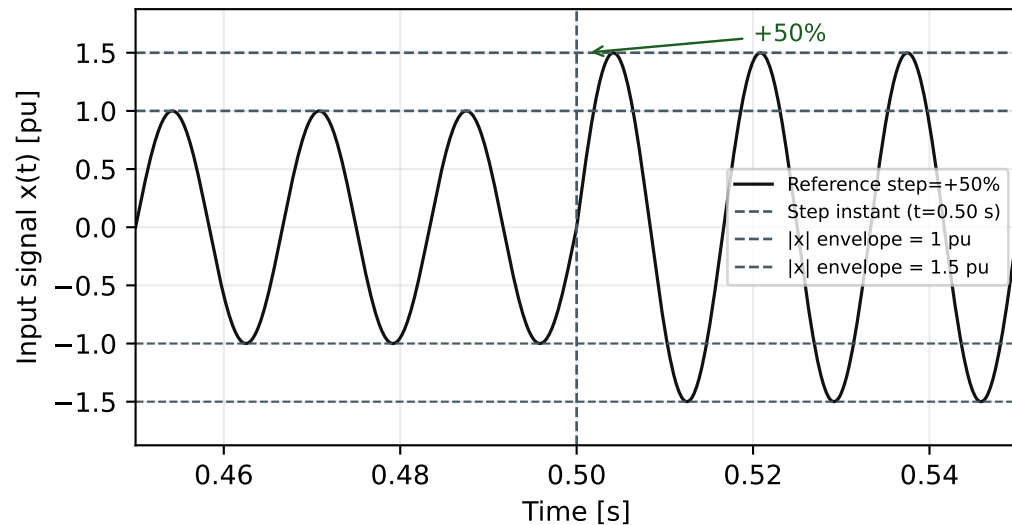
Adaptive



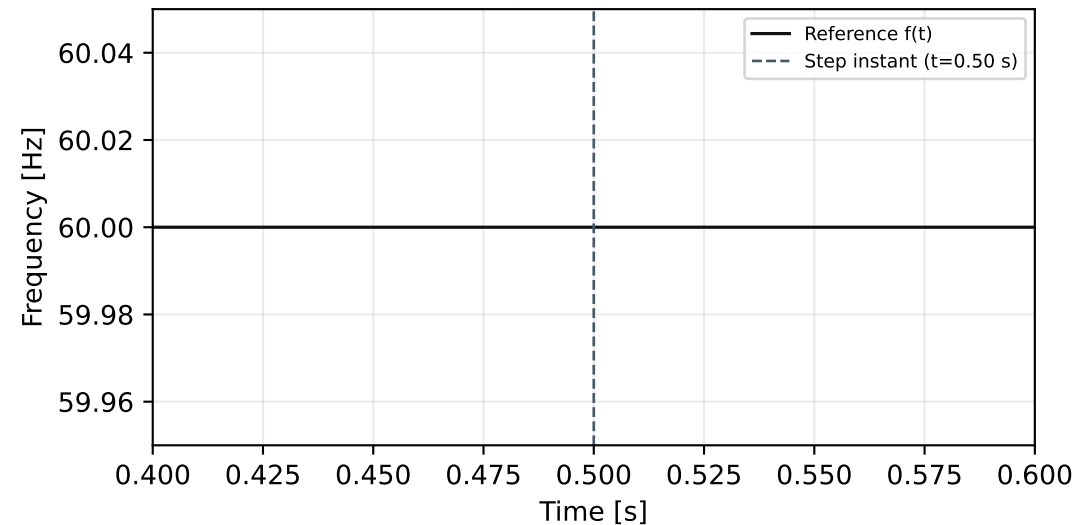
Runtime jitter: by estimator family (Fixed Policy)

This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify magnitude-step cross-sensitivity, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits. Fixed-policy mode: one parameter set per estimator is reused across all amplitudes.

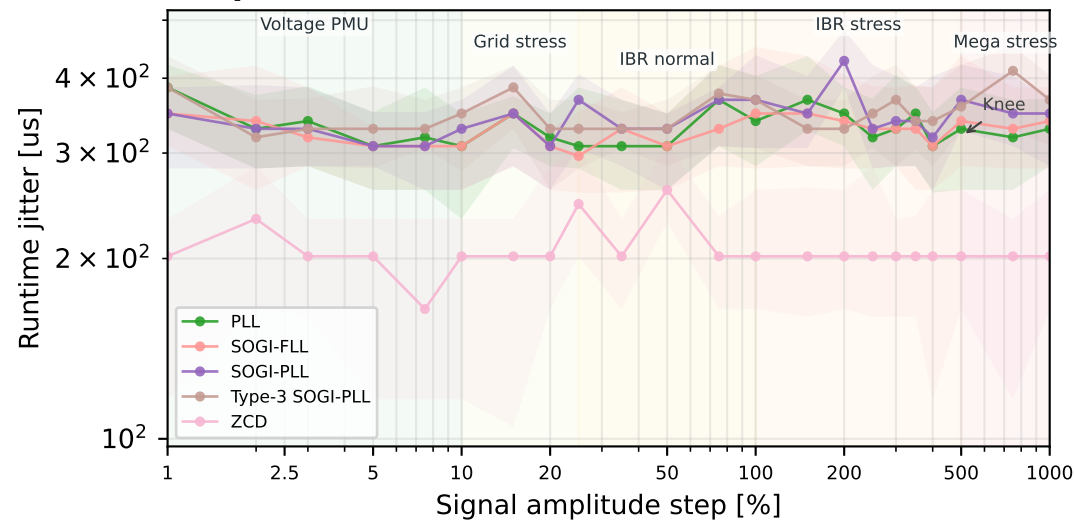
Reference Amplitude Step



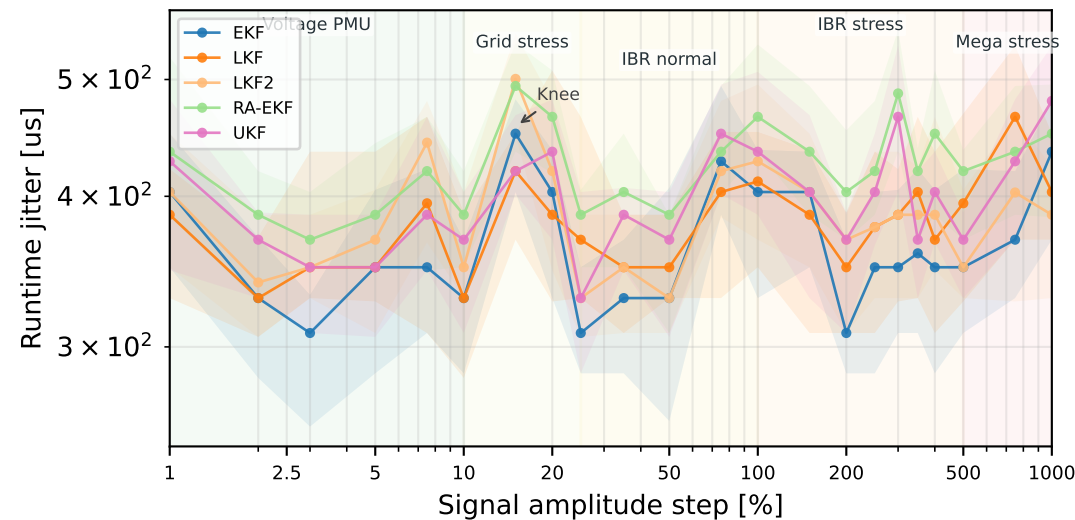
Reference Frequency $f(t)$



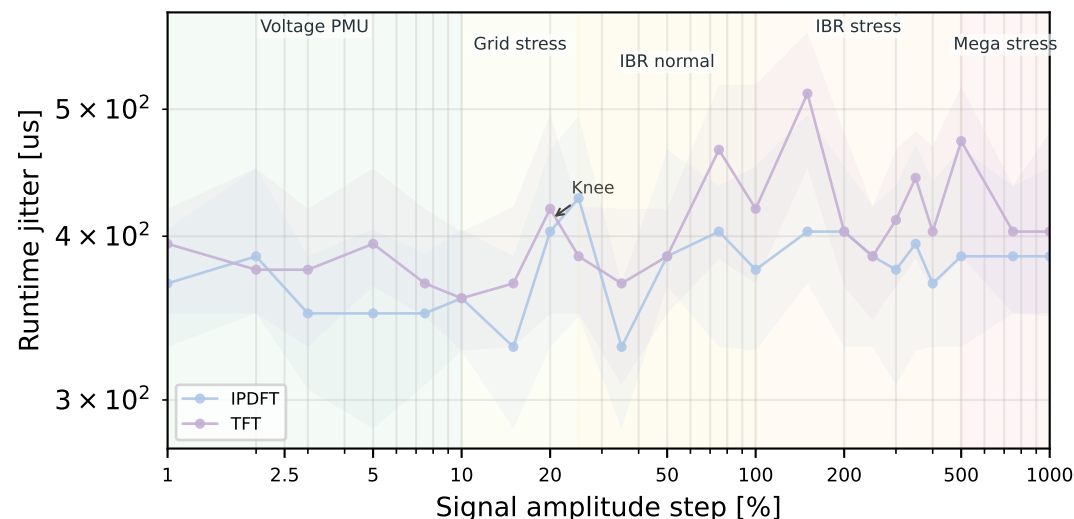
Loop-based



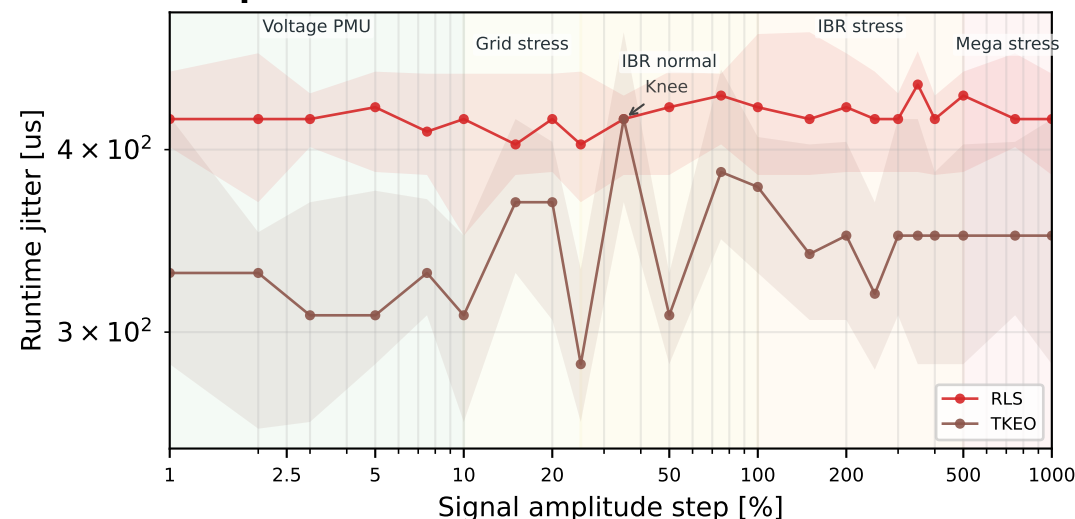
Model-based



Window-based



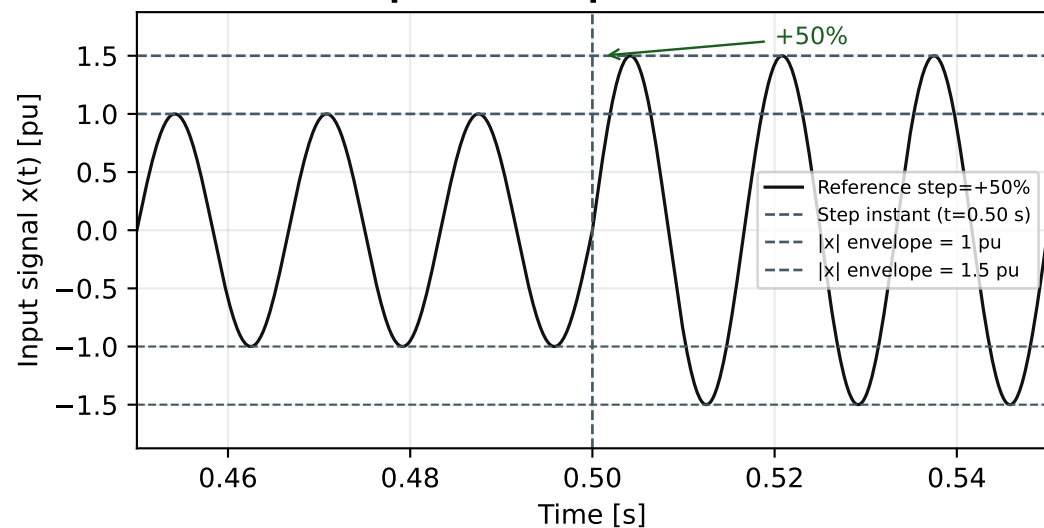
Adaptive



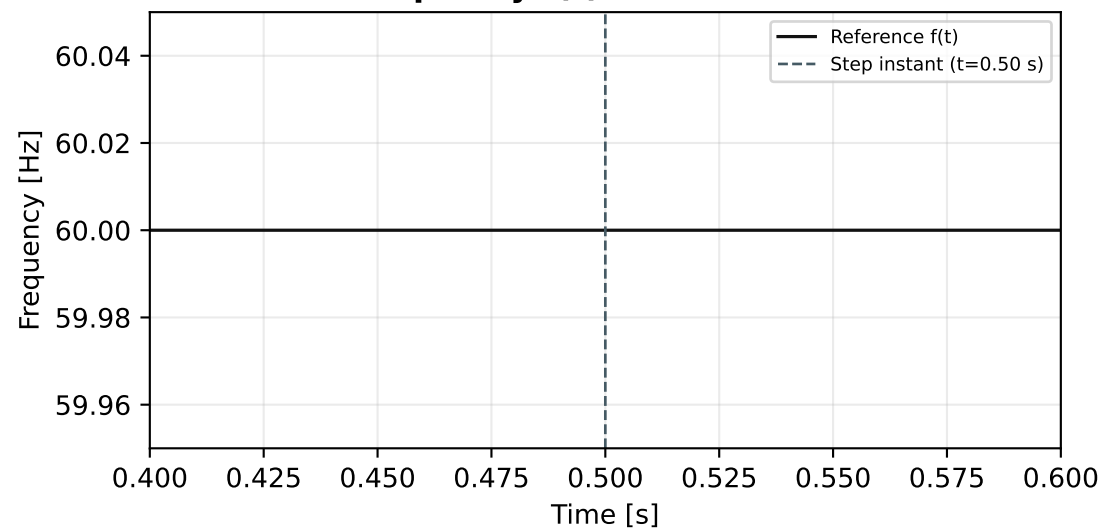
Invalid outputs: by estimator family (Fixed Policy)

This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify magnitude-step cross-sensitivity, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits. Fixed-policy mode: one parameter set per estimator is reused across all amplitudes.

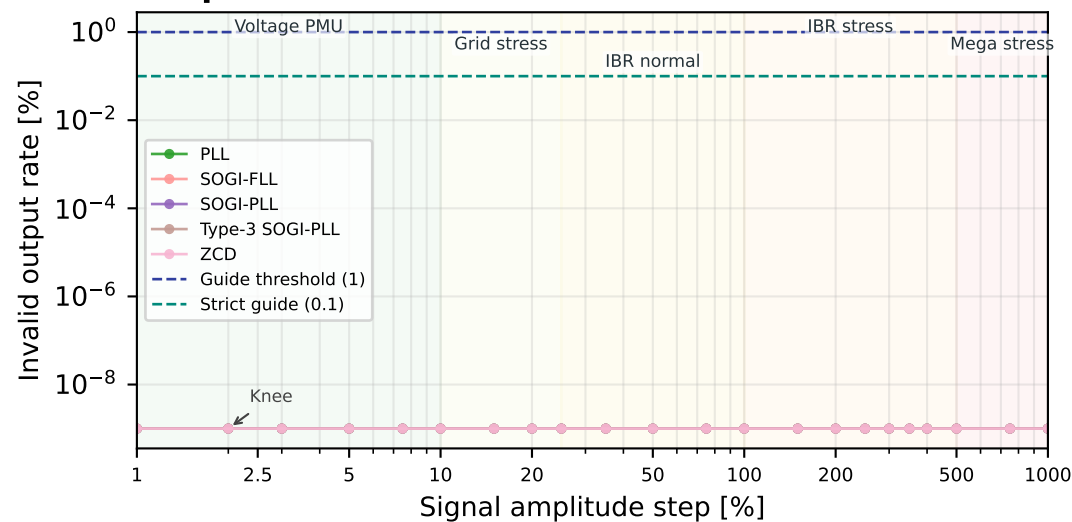
Reference Amplitude Step



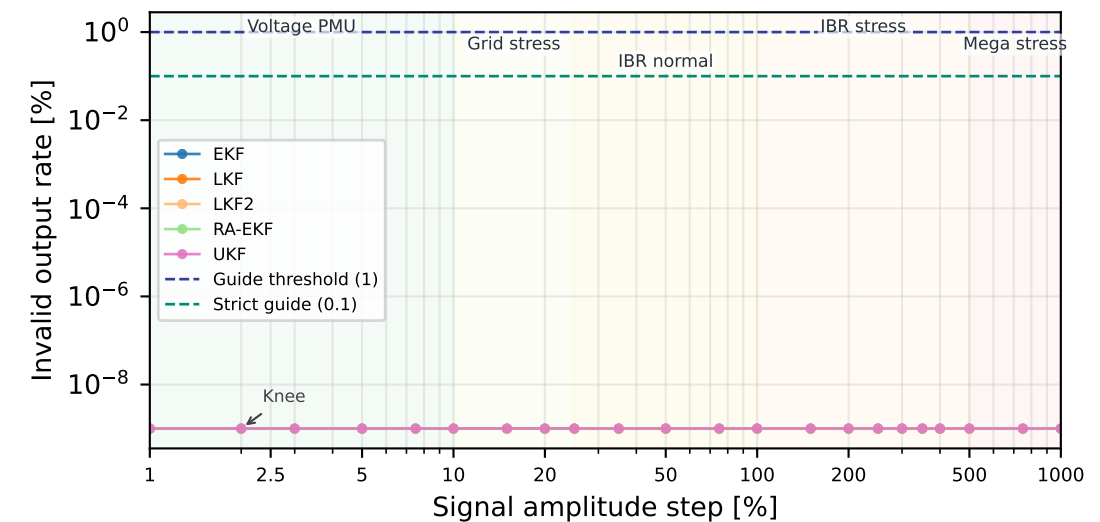
Reference Frequency f(t)



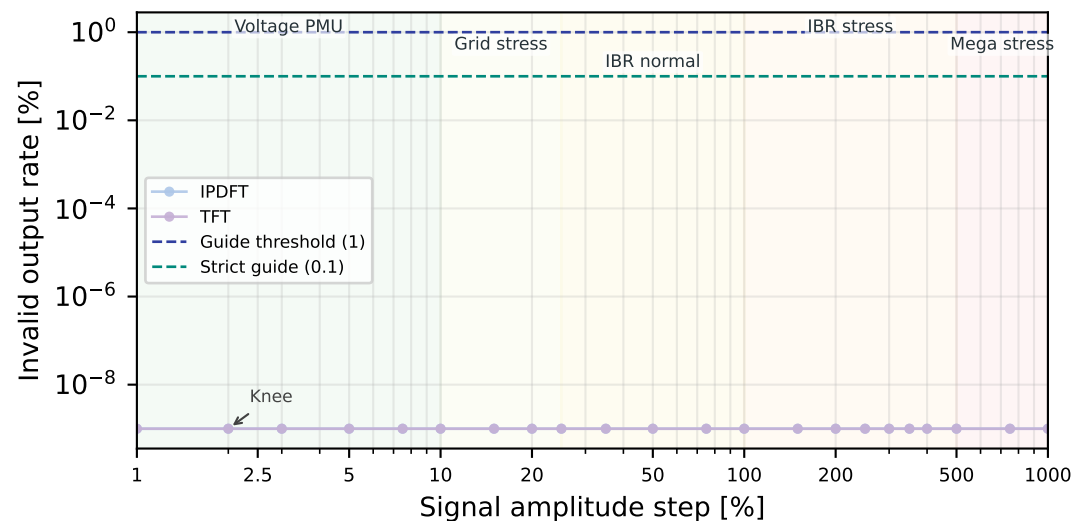
Loop-based



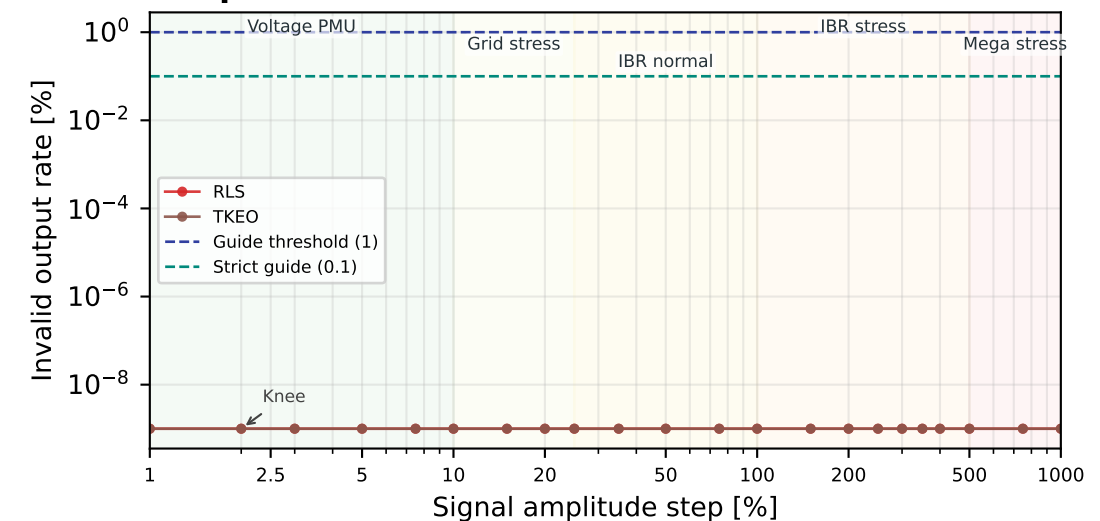
Model-based



Window-based



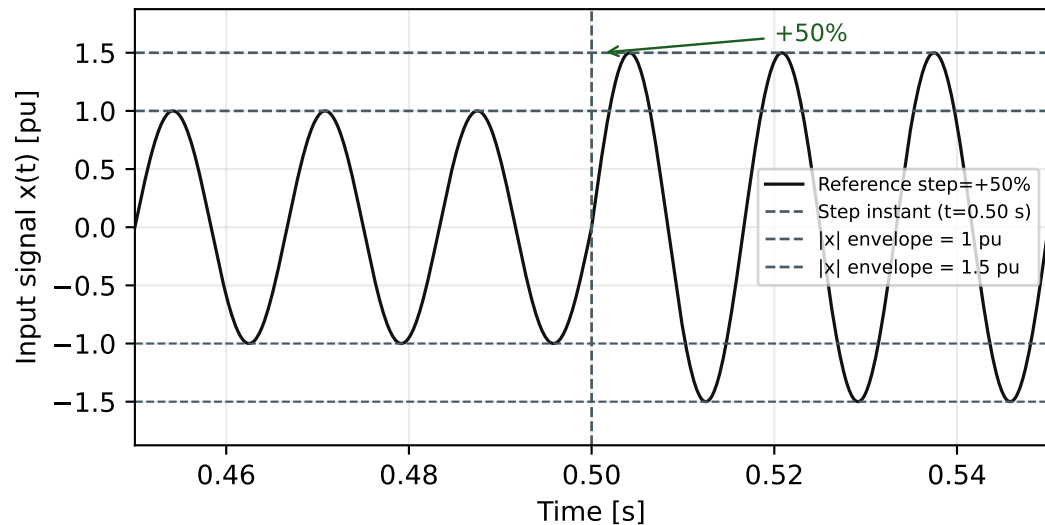
Adaptive



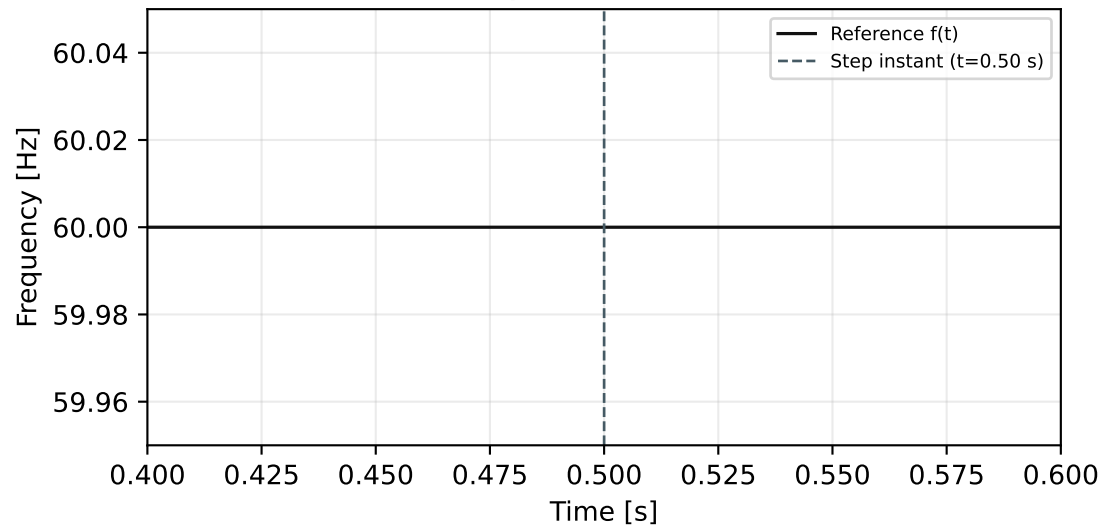
Startup validity: by estimator family (Fixed Policy)

This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify magnitude-step cross-sensitivity, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits. Fixed-policy mode: one parameter set per estimator is reused across all amplitudes.

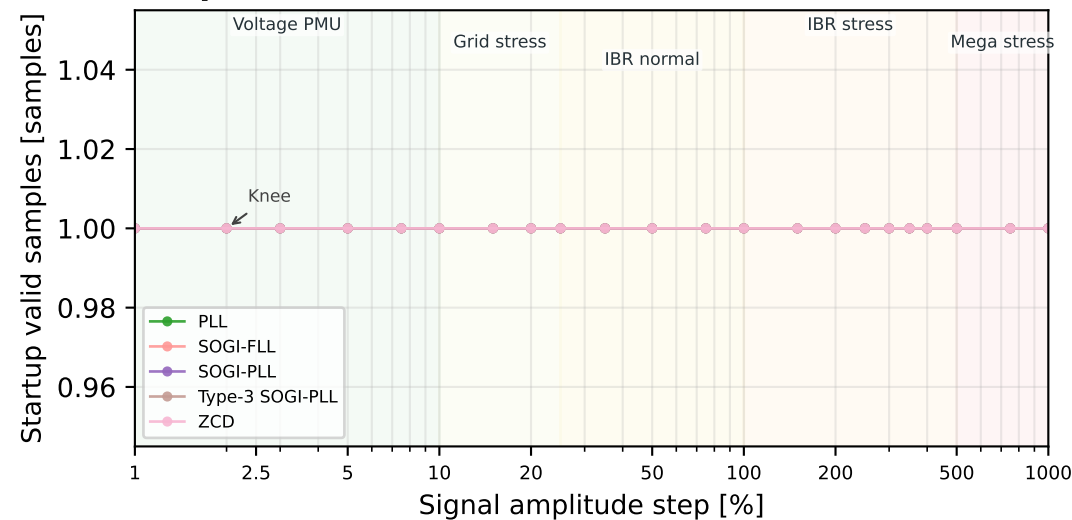
Reference Amplitude Step



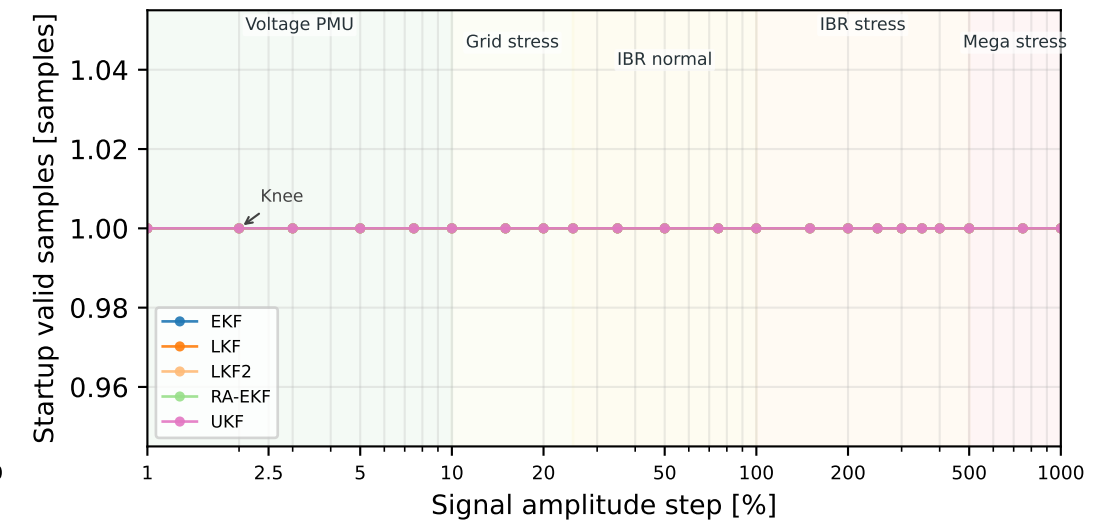
Reference Frequency $f(t)$



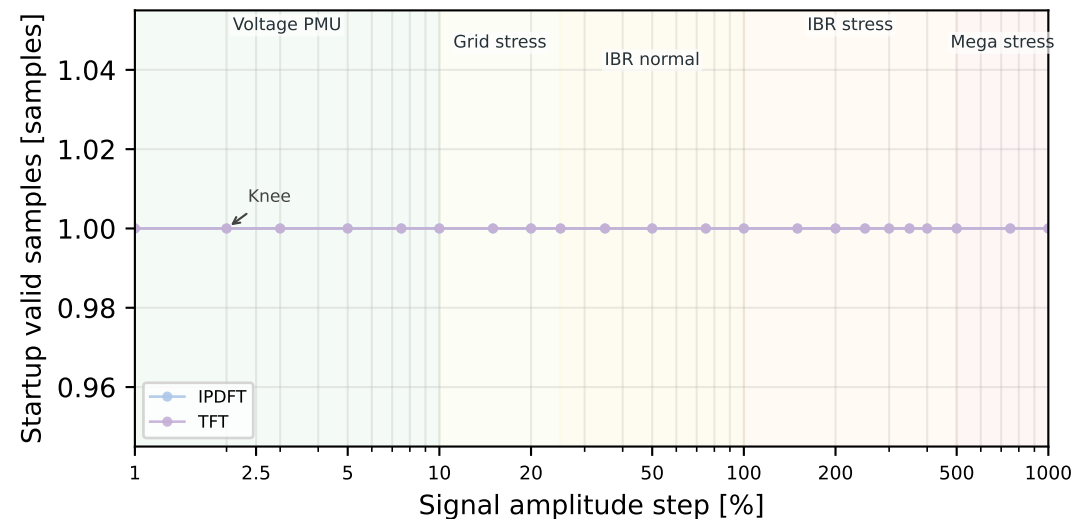
Loop-based



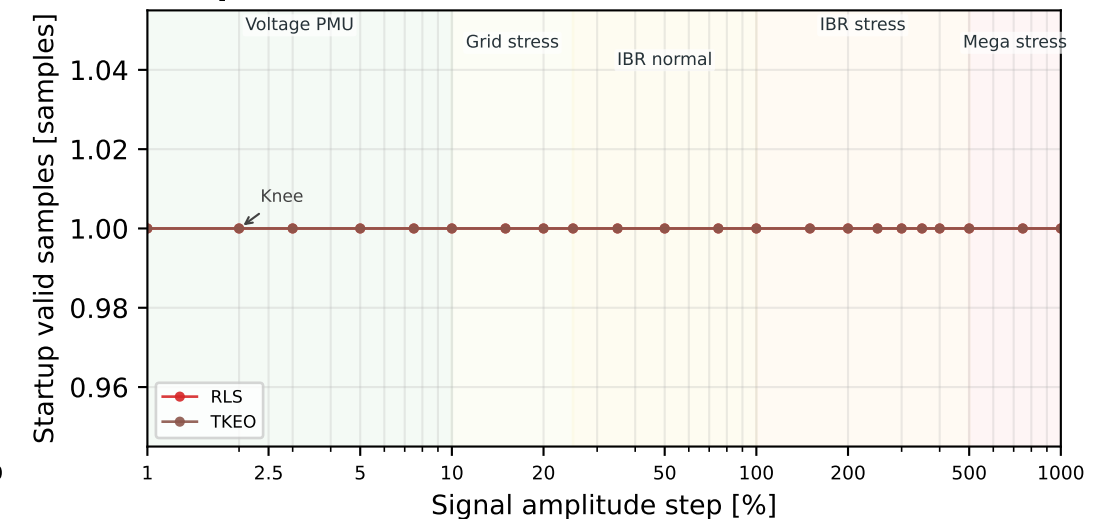
Model-based



Window-based



Adaptive



Phase Dispersion of Sensitive Estimators (Fixed Policy)

Each marker is one Monte Carlo run. This page exposes phase-dependent AM-to-FM leakage that is hidden by median-only curves.

